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Soo et al.

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(54) **INTERSPINOUS-INTERLAMINAR
STABILIZATION SYSTEMS AND METHODS**

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A61B 17/70 (2006.01)

(52) **U.S. Cl.**
CPC **A61B 17/7067** (2013.01); **A61B 17/7065** (2013.01); **A61B 17/7071** (2013.01)

(58) **Field of Classification Search**

CPC A61B 17/7065; A61B 17/7067; A61B 17/7071; A61F 2/4611

See application file for complete search history.

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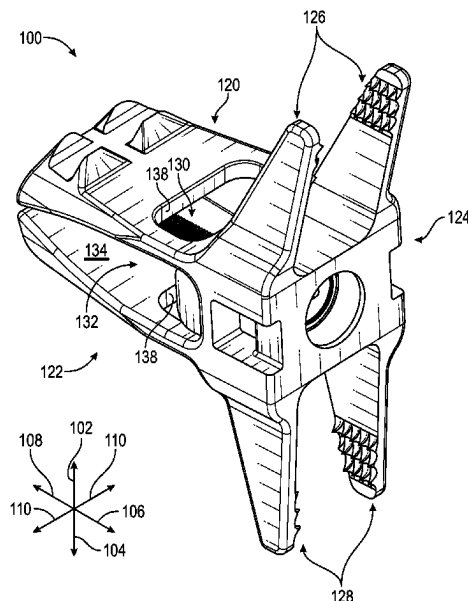
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Maywood IP Law

(57) **ABSTRACT**

Systems and methods are disclosed for maintaining spacing between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina of adjacent vertebrae of a spine. A system may include an implant having a proximal superior surface with a superior concavity shaped to receive the superior spinous process, a proximal inferior surface with an inferior concavity shaped to receive the inferior spinous process, a distal superior surface, distal to the proximal superior surface, that faces the superior lamina, and a distal inferior surface, distal to the proximal inferior surface, that faces the inferior lamina. The implant may further have a threaded member extending along a proximal-distal direction, that rotates to urge the implant to move from a retracted configuration to a deployed configuration by urging the distal superior surface and the distal inferior surface to move apart.

18 Claims, 22 Drawing Sheets



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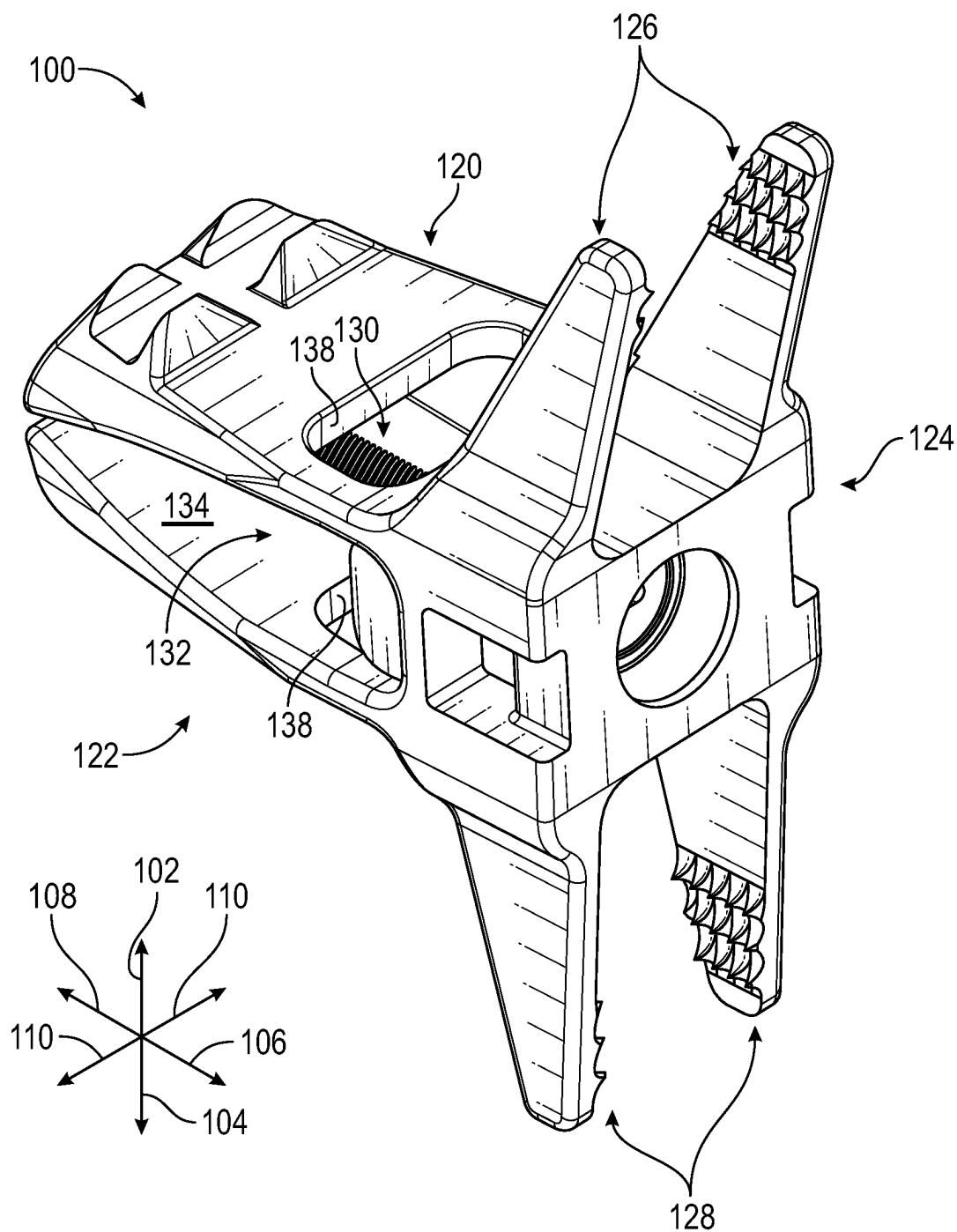


FIG. 1A

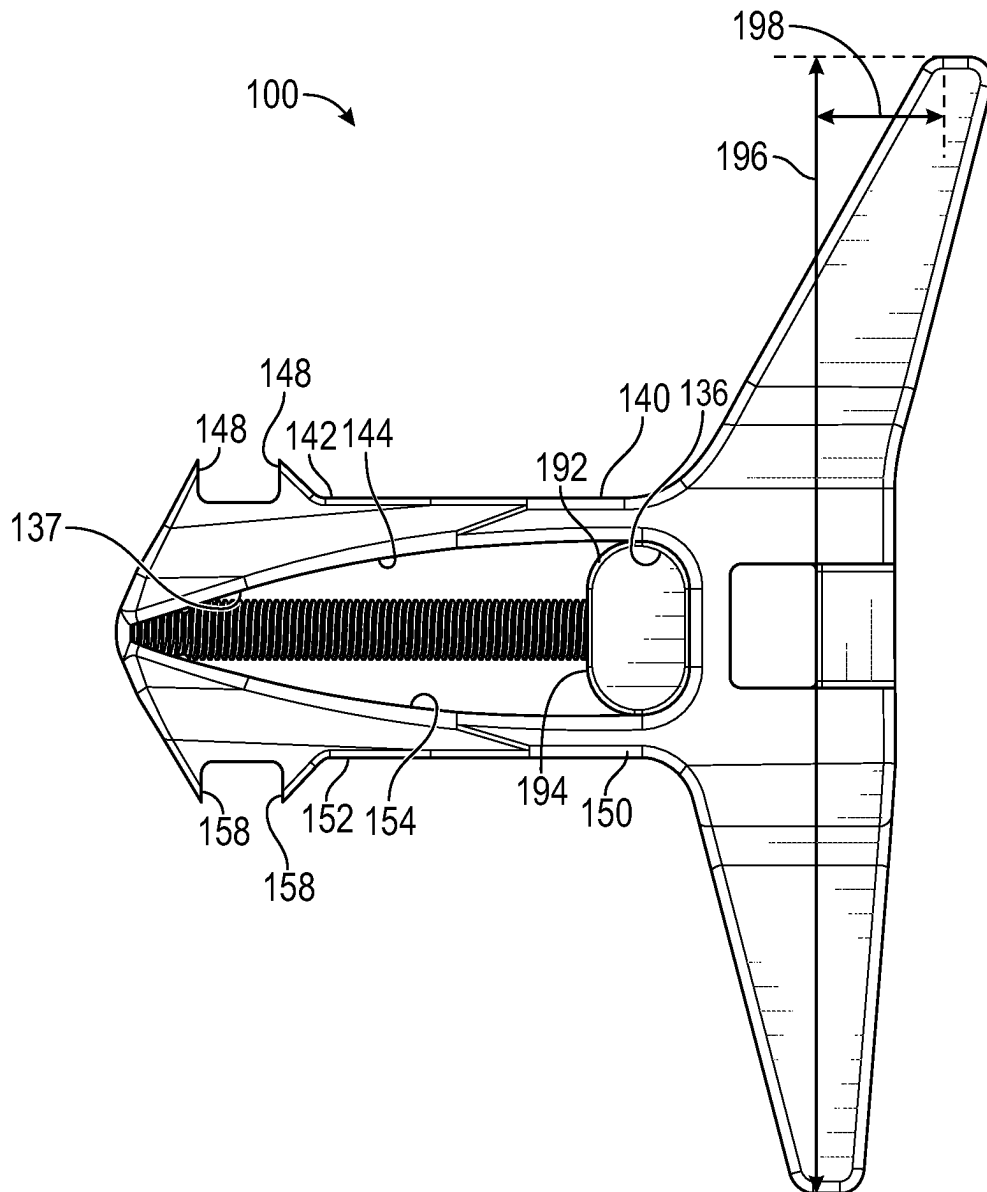


FIG. 1B

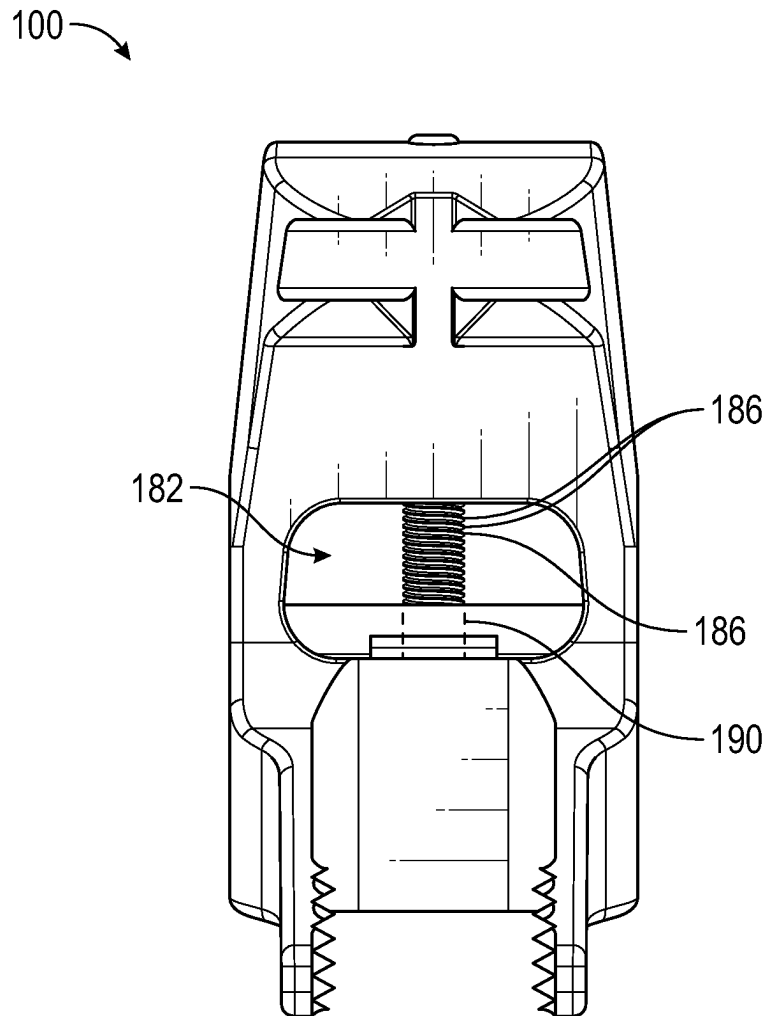


FIG. 1C

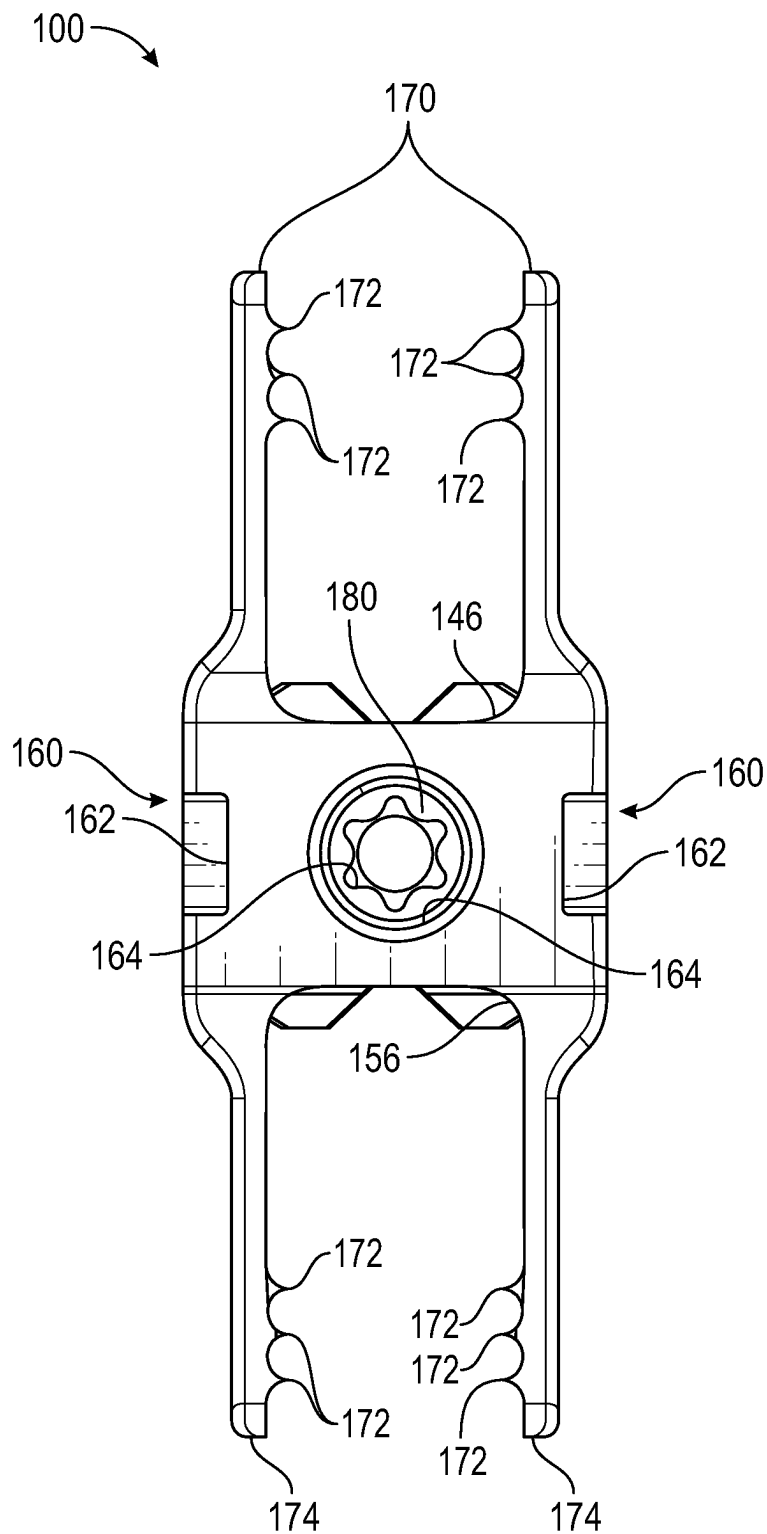


FIG. 1D

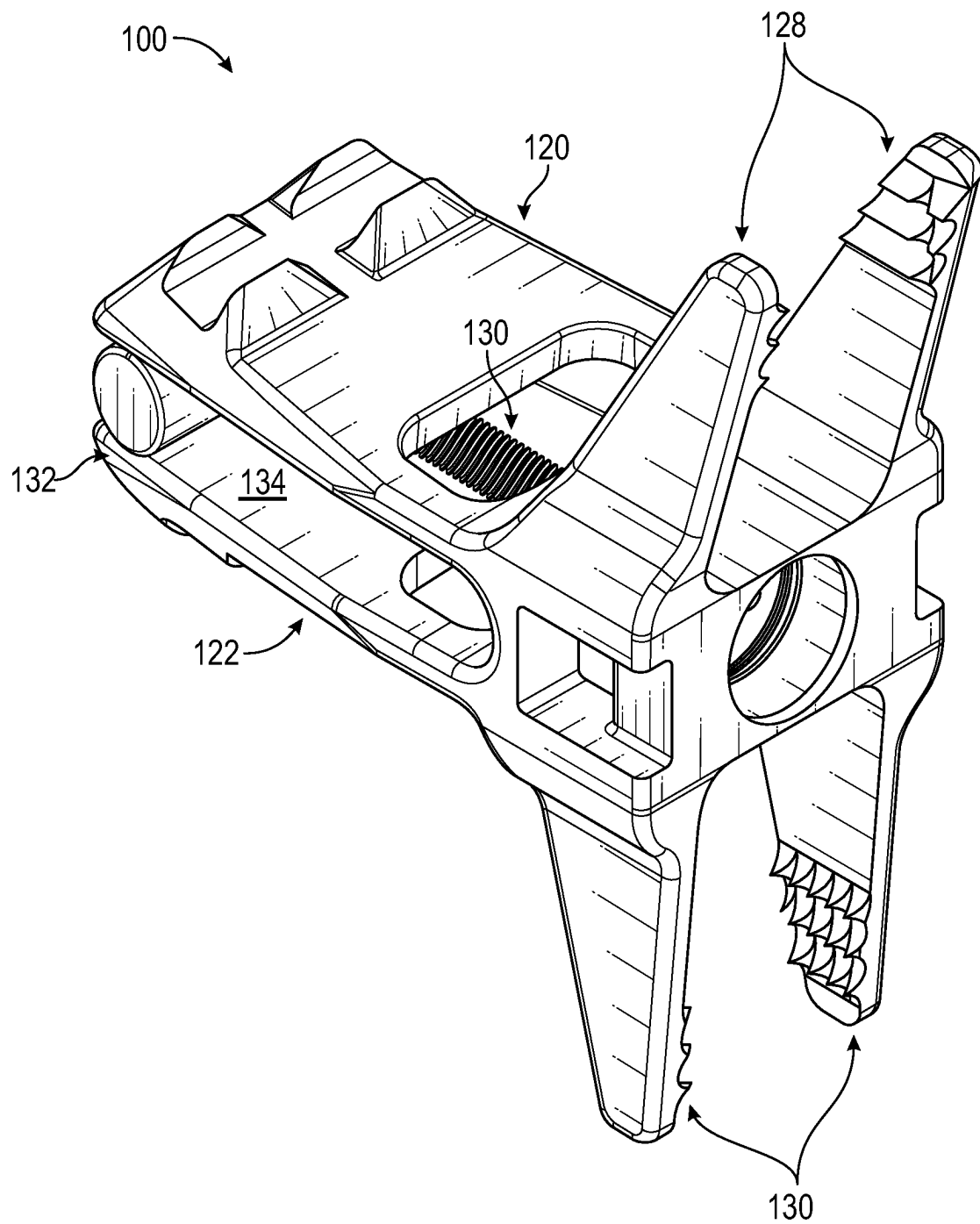


FIG. 2A

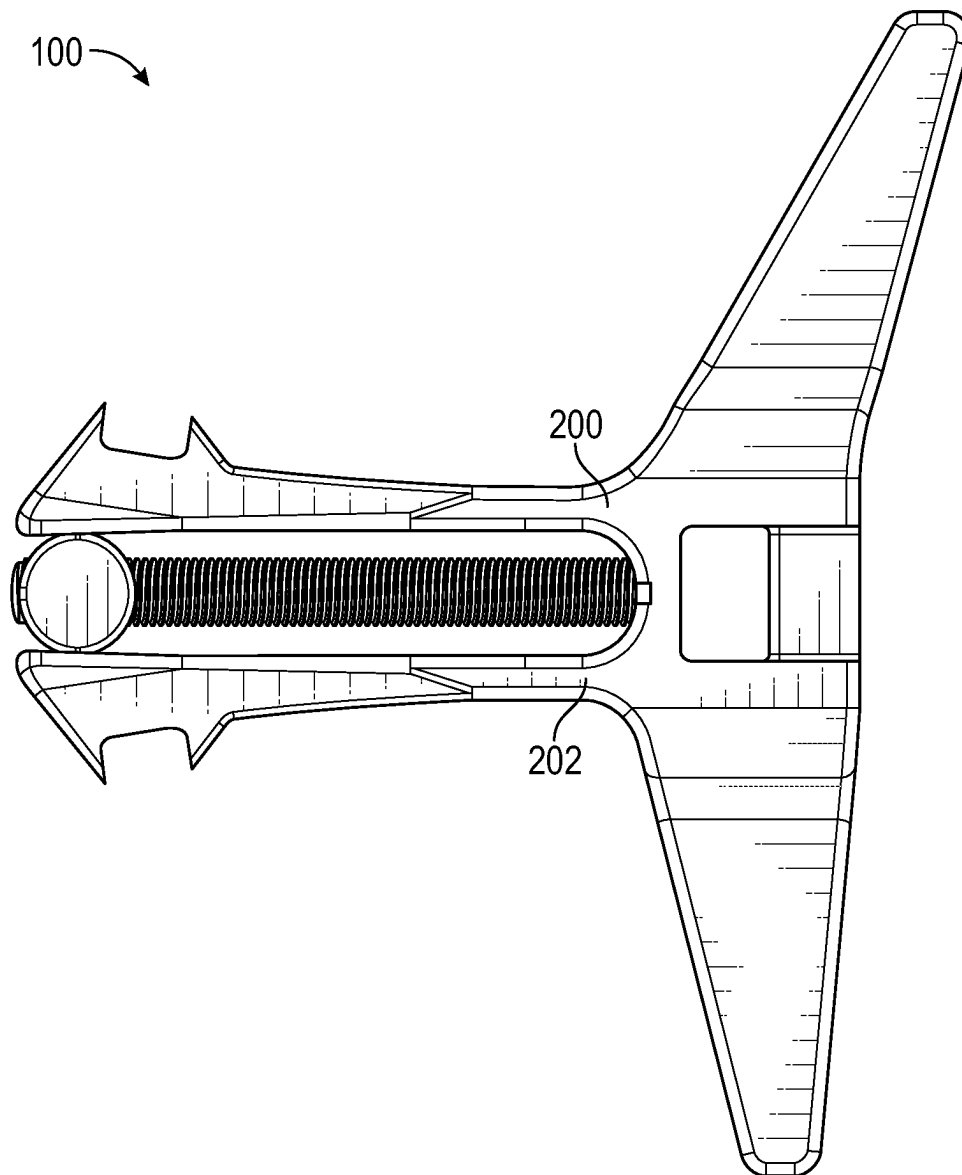


FIG. 2B

100 →

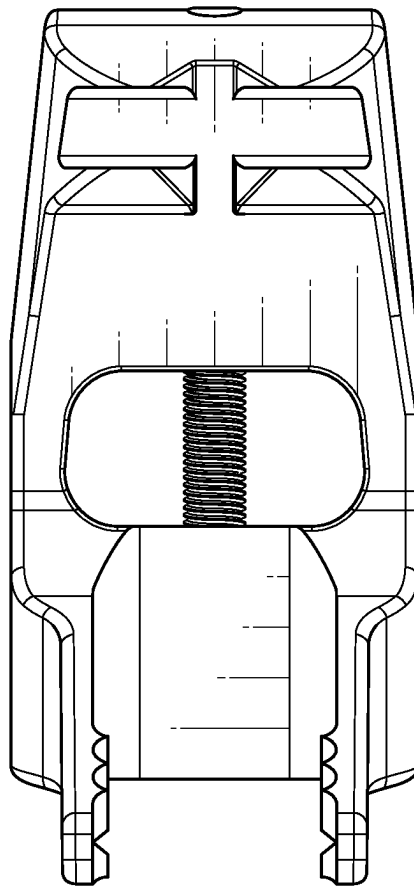


FIG. 2C

100 →

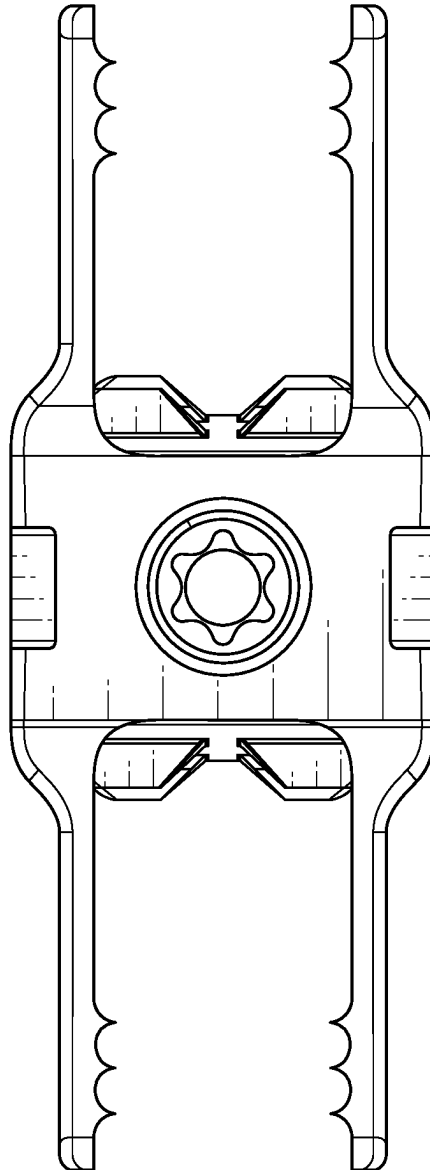


FIG. 2D

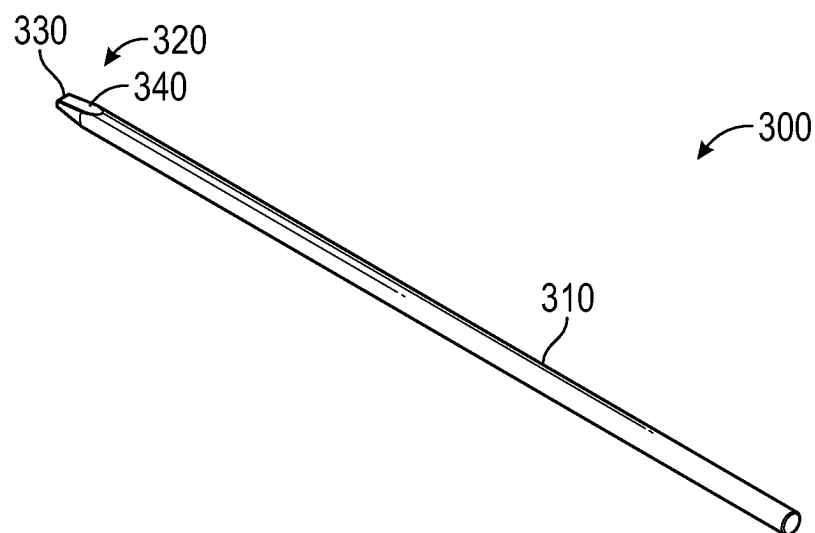


FIG. 3A

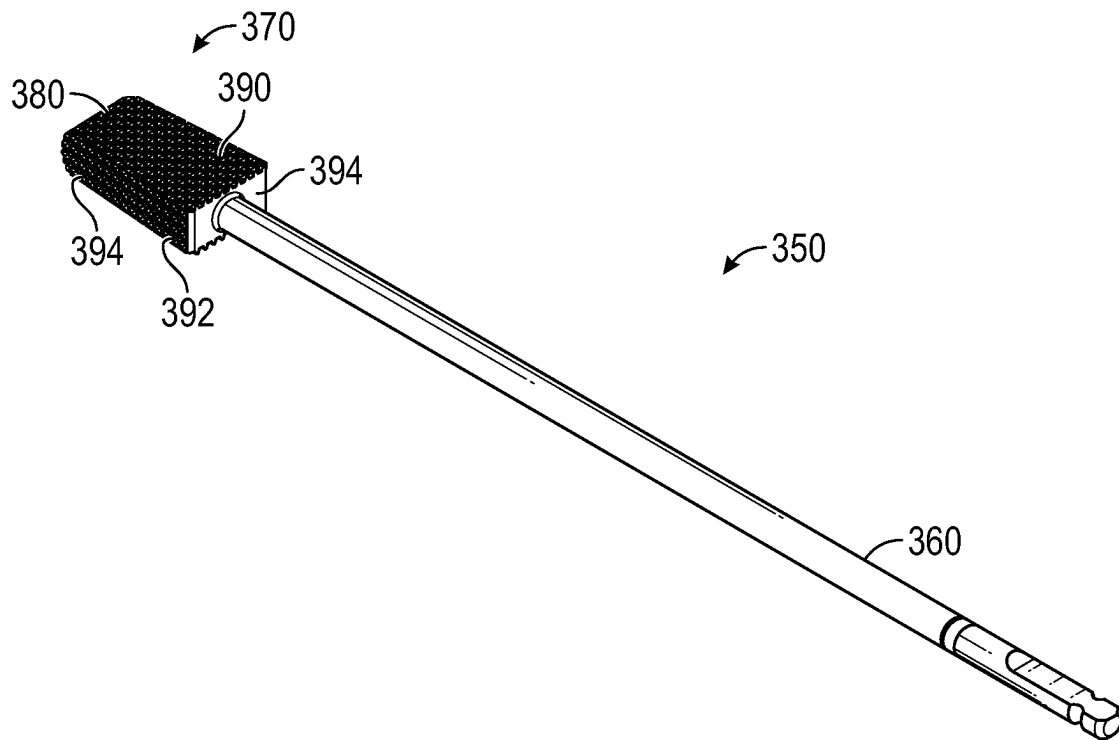


FIG. 3B

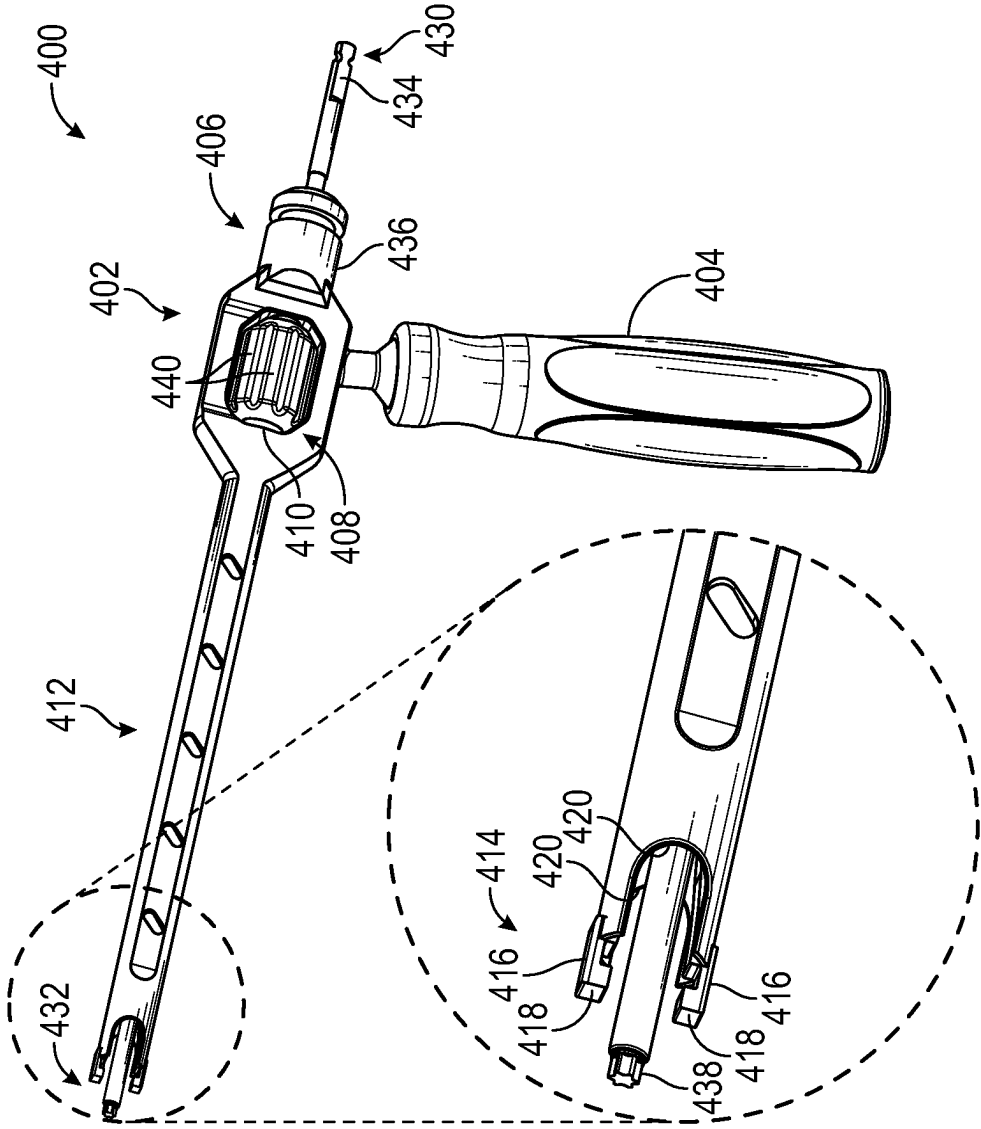


FIG. 4

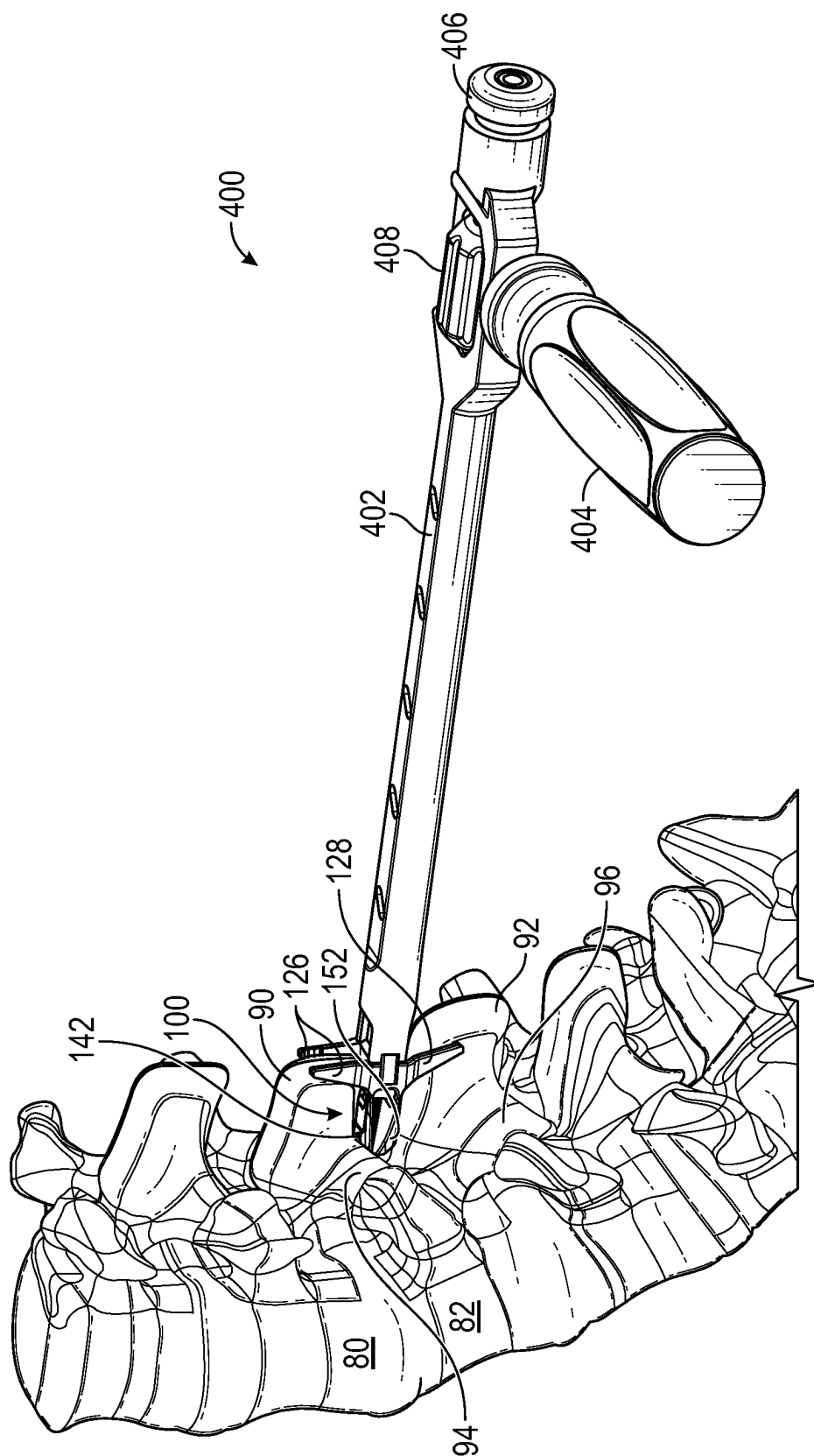


FIG. 5A

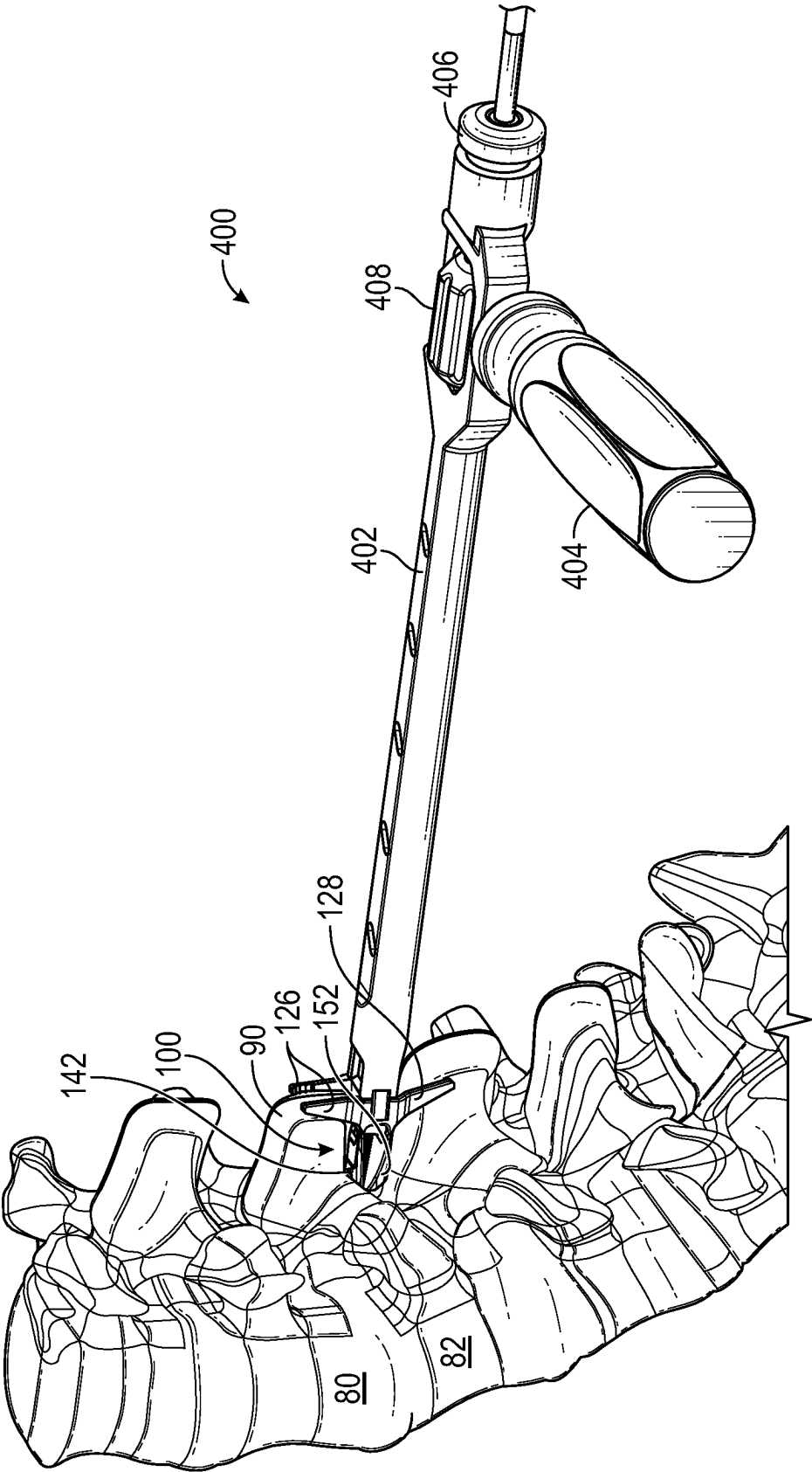


FIG. 5B

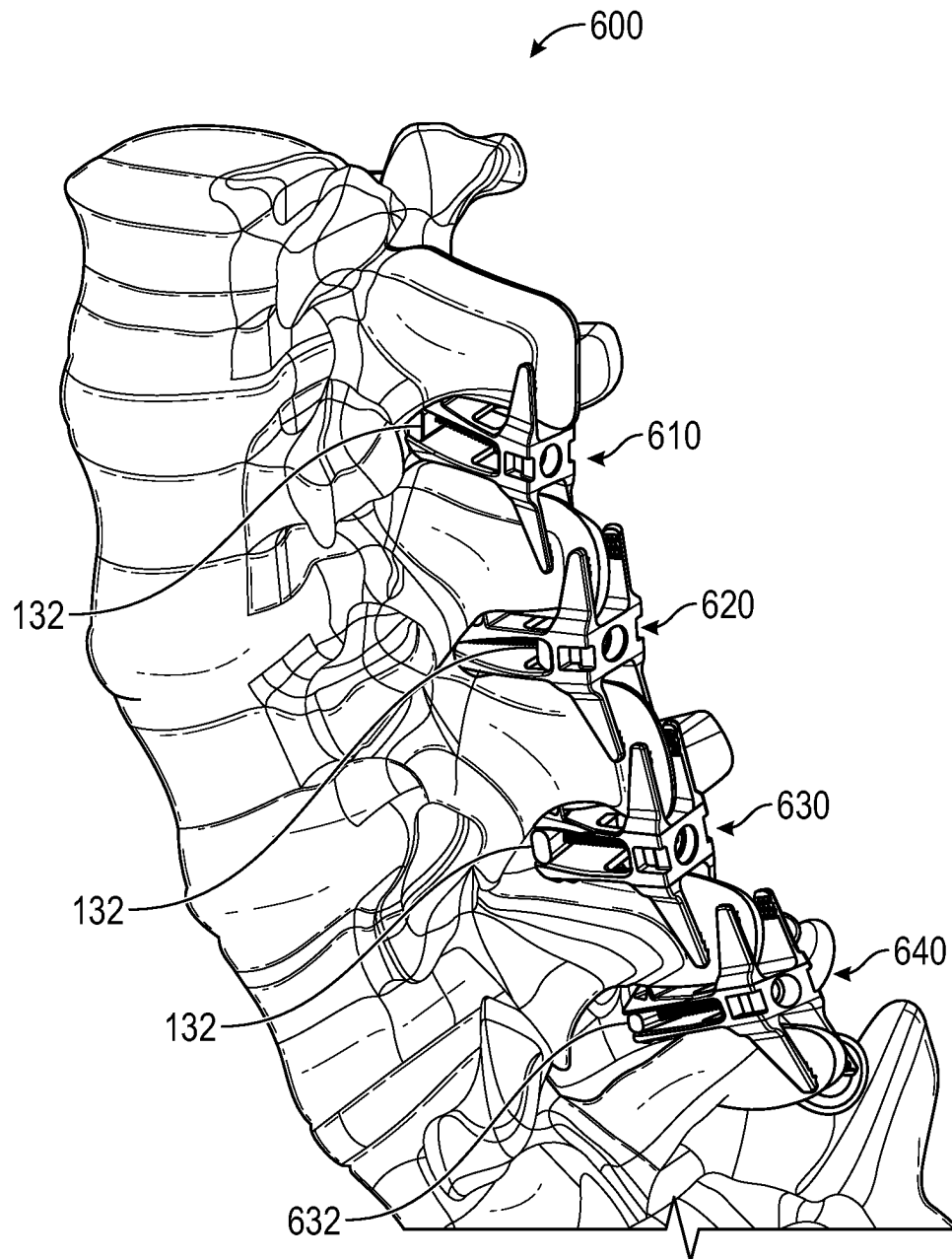


FIG. 6

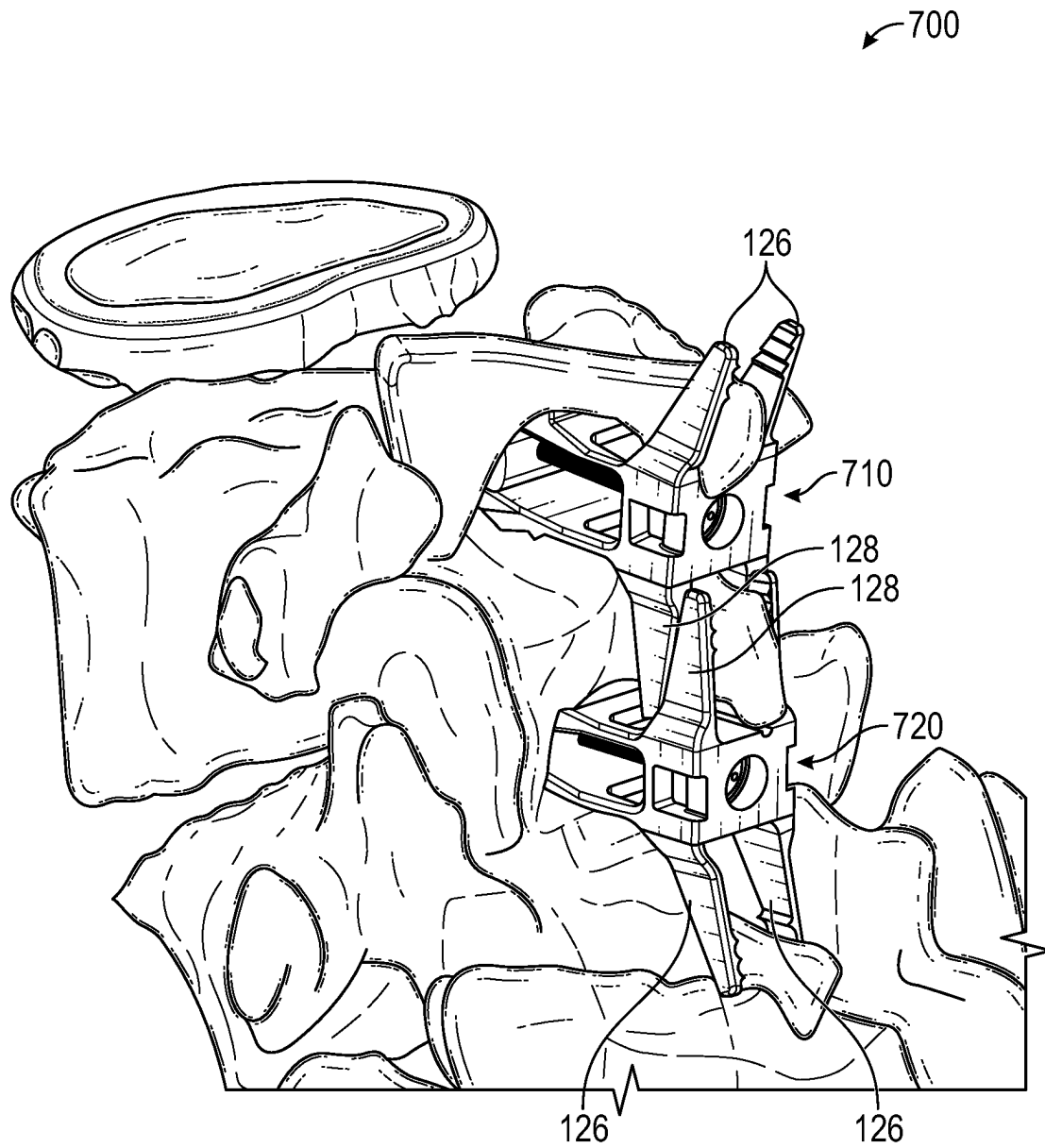


FIG. 7

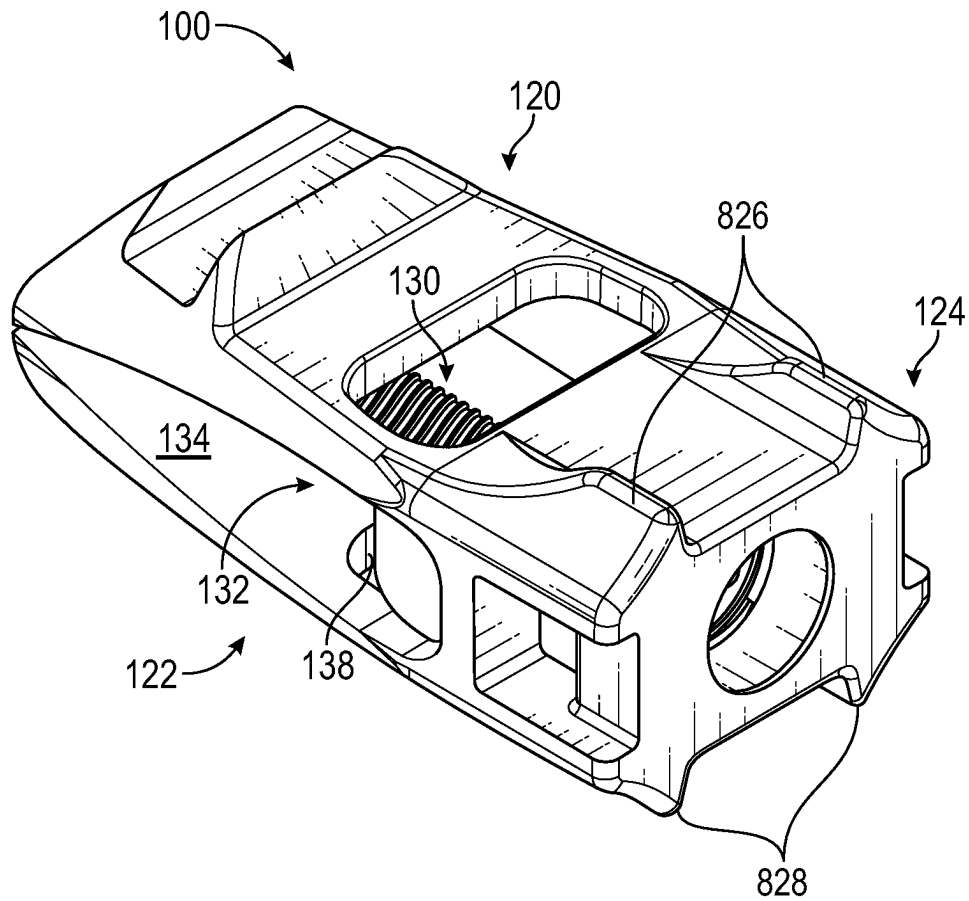


FIG. 8A

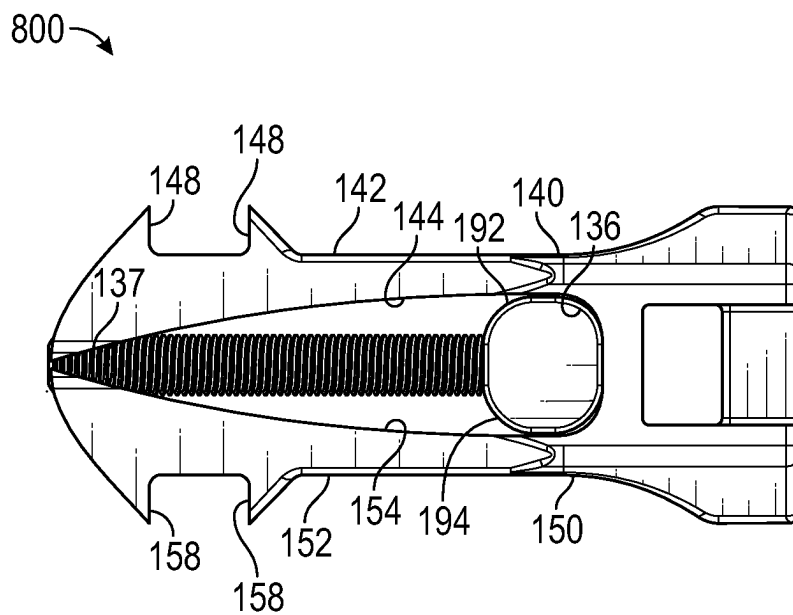


FIG. 8B

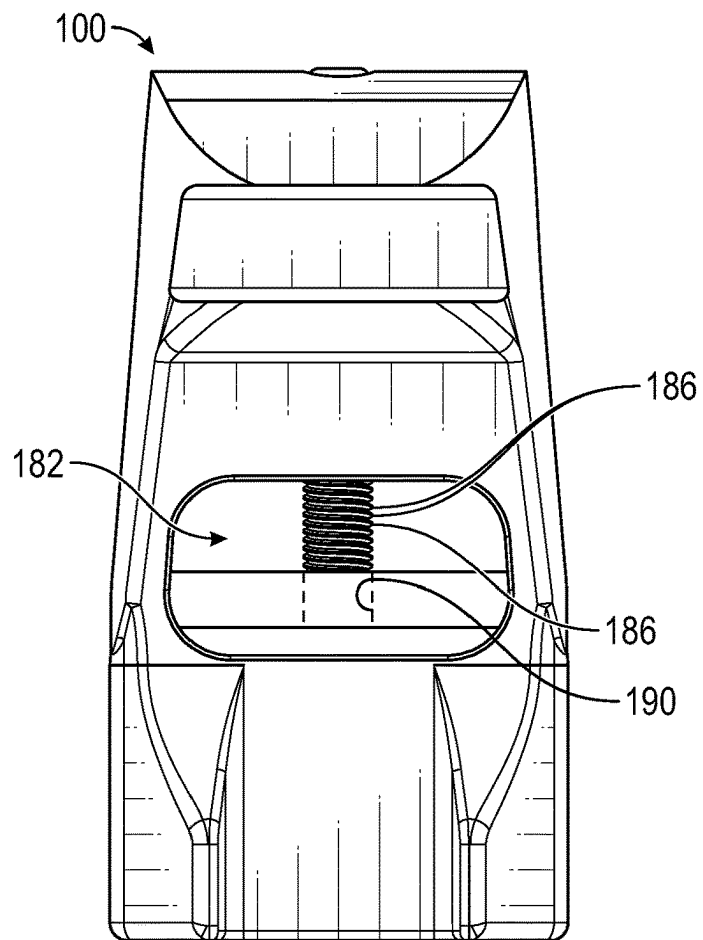


FIG. 8C

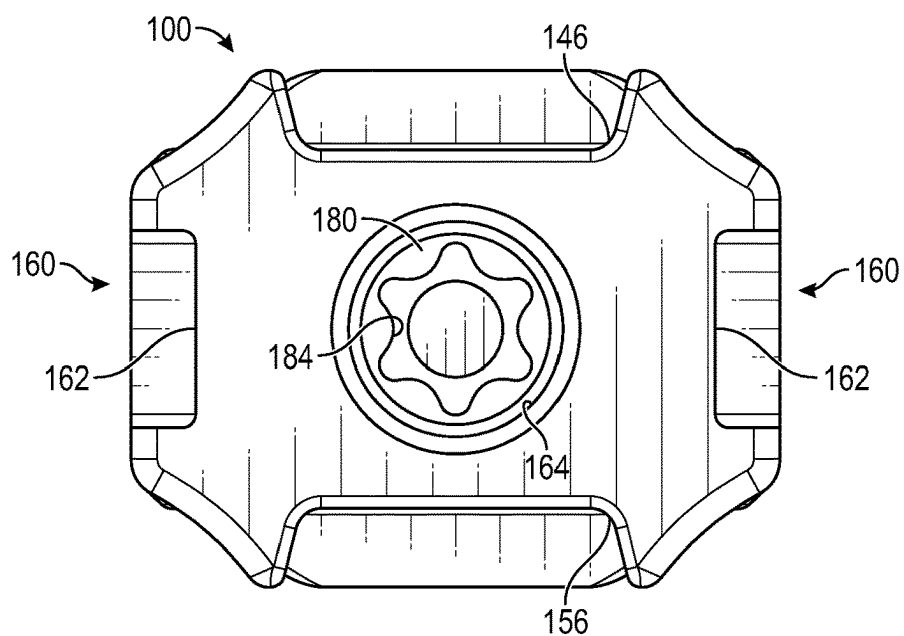


FIG. 8D

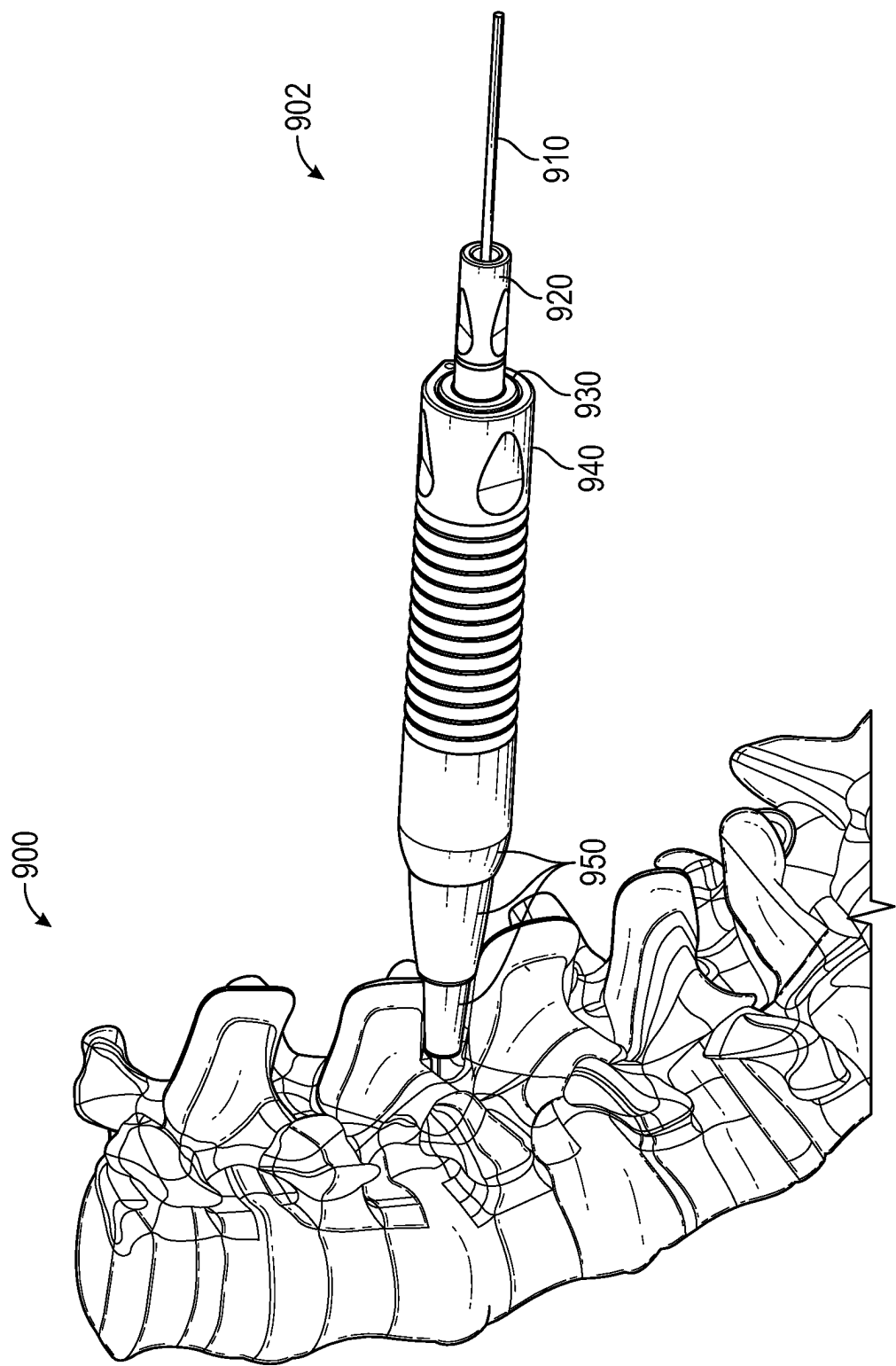


FIG. 9A

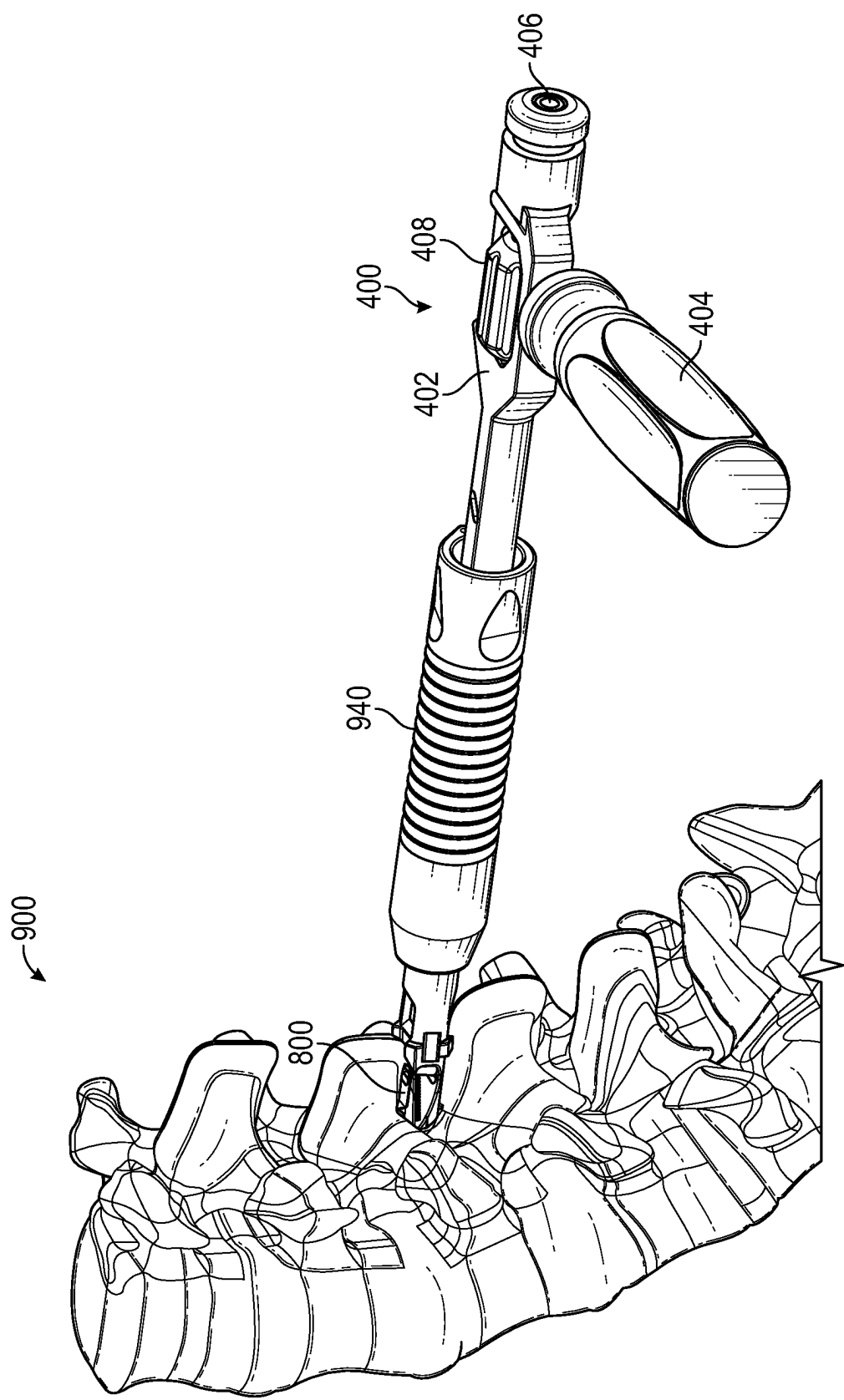


FIG. 9B

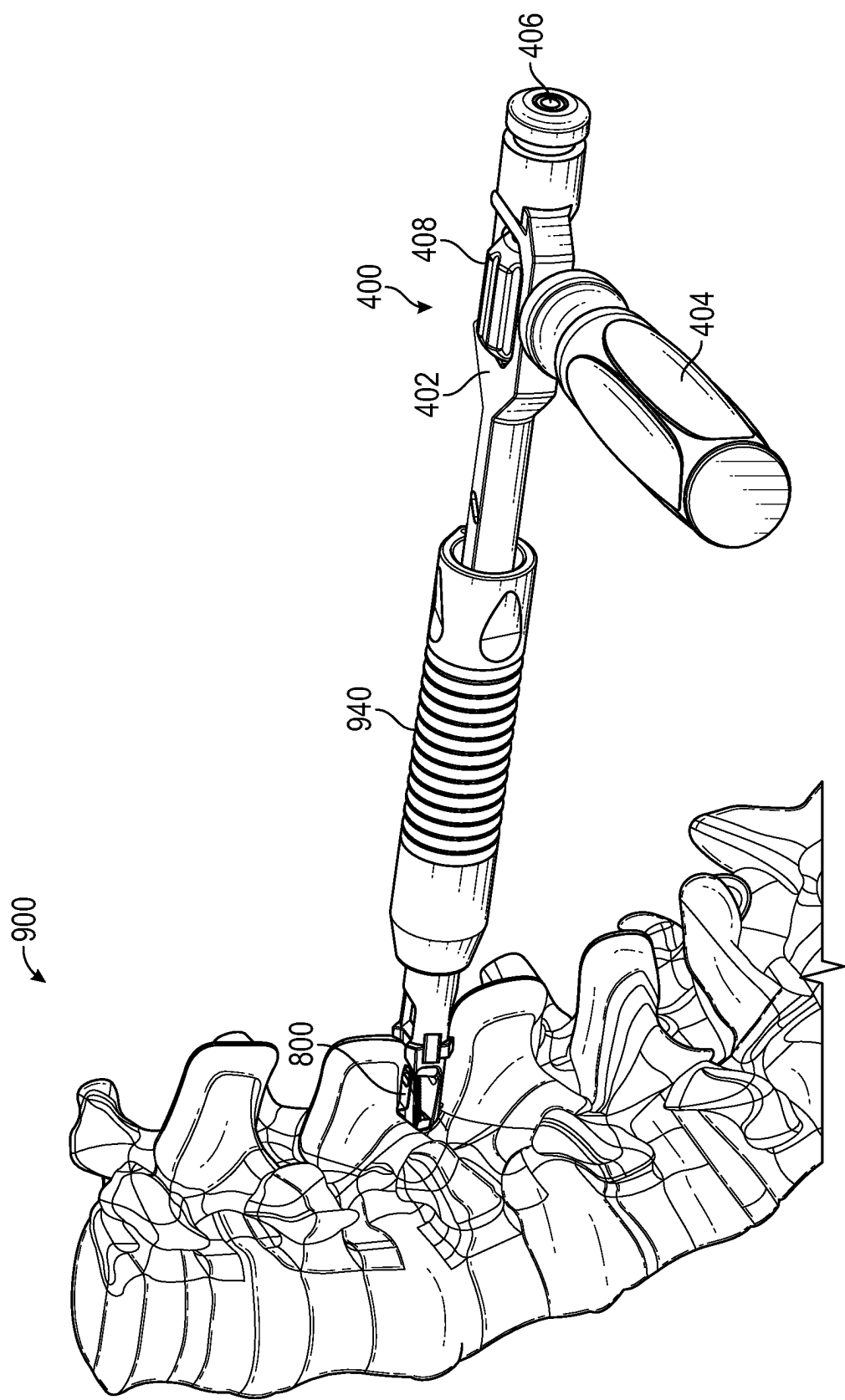


FIG. 10A

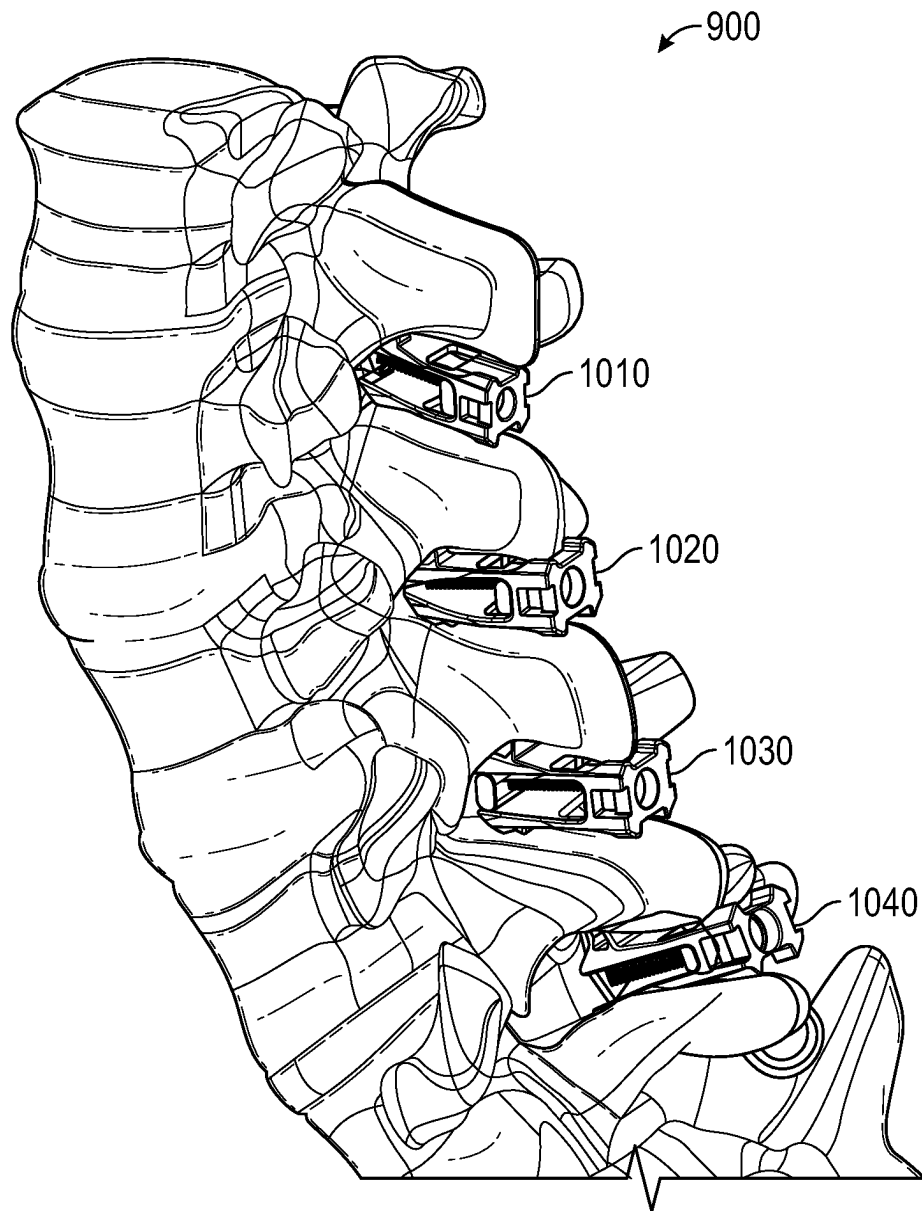


FIG. 10B

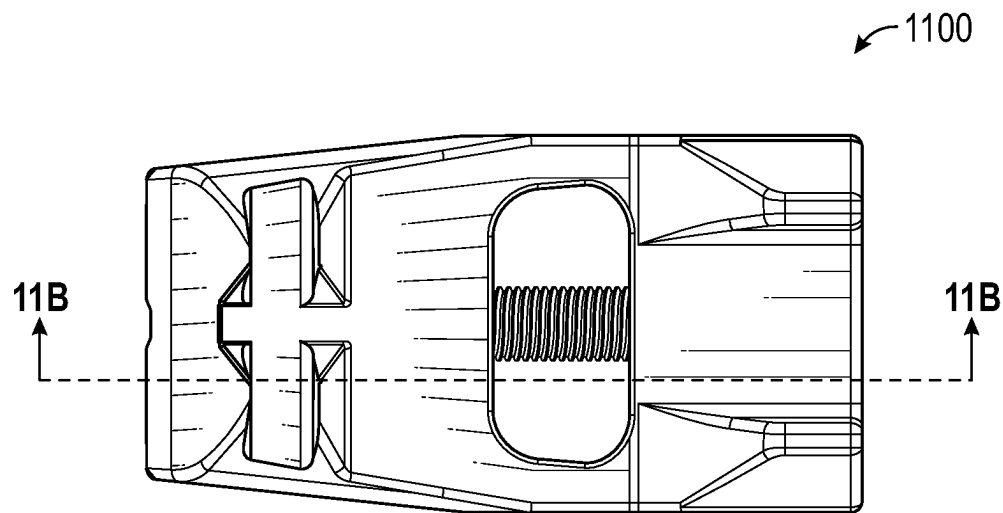


FIG. 11A

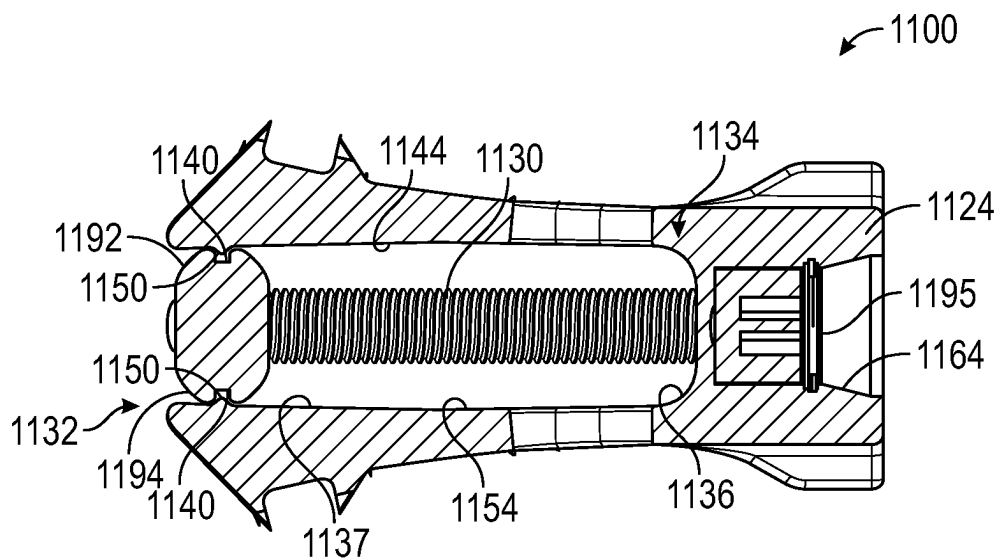


FIG. 11B

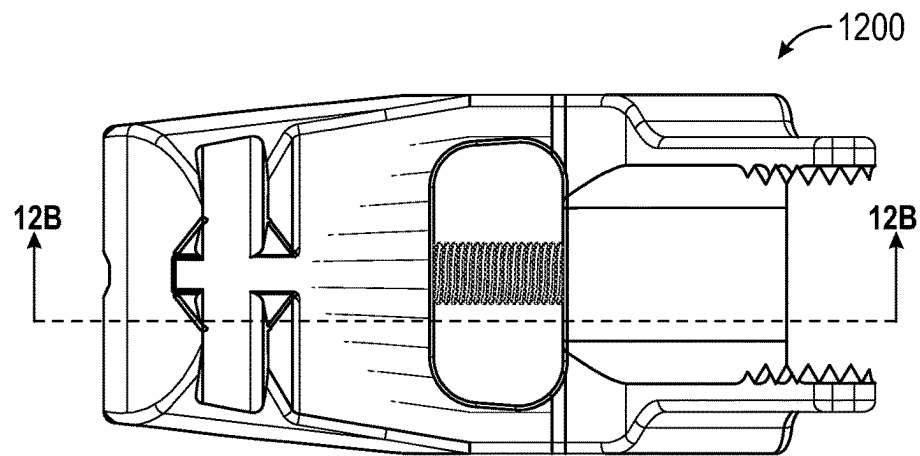


FIG. 12A

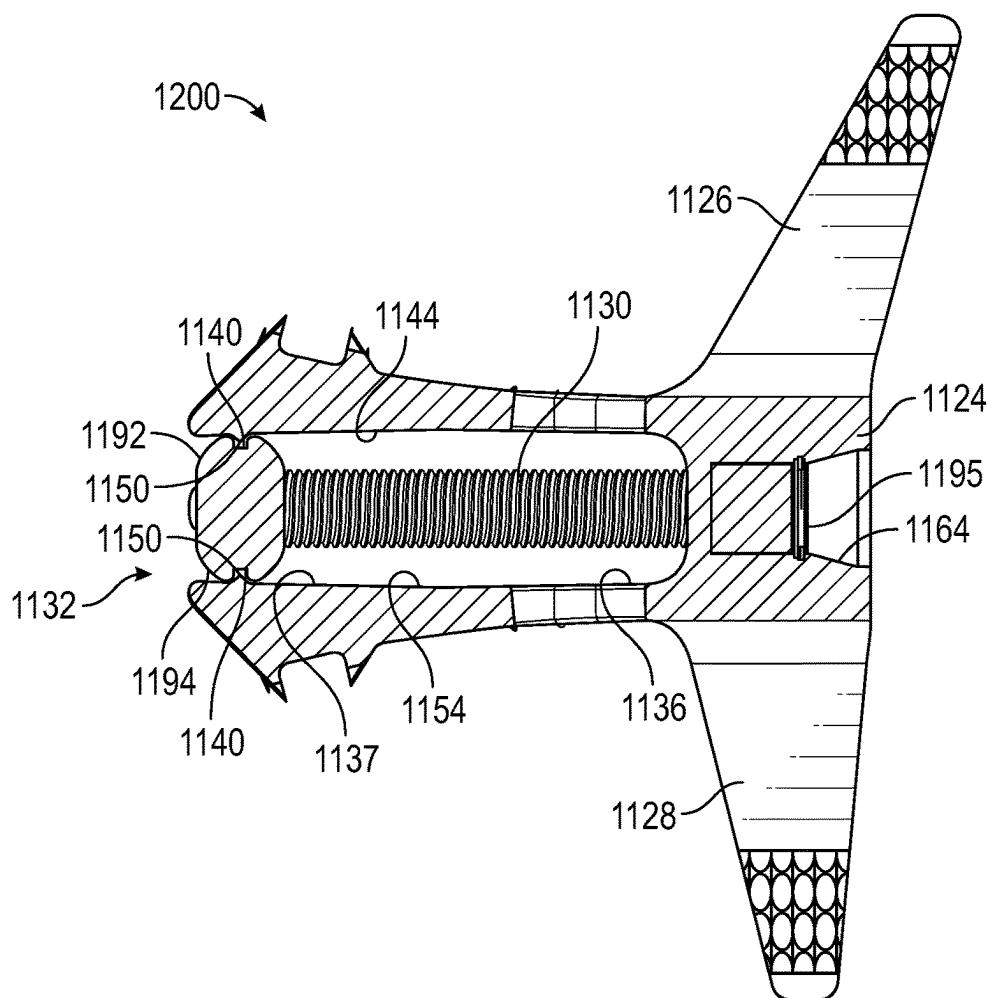


FIG. 12B

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INTERSPINOUS-INTERLAMINAR STABILIZATION SYSTEMS AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/732,823 filed on Apr. 29, 2022, which is a divisional of U.S. patent application Ser. No. 17/000,033 filed on Aug. 21, 2020, entitled INTERSPINOUS-INTERLAMINAR STABILIZATION SYSTEMS AND METHODS, which issued on May 3, 2022 as U.S. Pat. No. 11,317,950, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/889,719, entitled INTERSPINOUS DISTRACTION DEVICE, which was filed on Aug. 21, 2019, which are incorporated by reference as though set forth herein in their entirety.

TECHNICAL FIELD

The present disclosure relates generally to surgical systems and methods, and more particularly, to systems and methods for maintaining a desired level of distraction between the spinous processes and laminae of adjacent vertebrae.

BACKGROUND

Various spinal conditions can cause instability in one or more levels of the posterior spine. In particular, narrowing of the spinal canal, known as spinal stenosis and degeneration of the intervertebral disc, can enable the posterior aspects of adjacent vertebrae to come together, which may cause pain or damage by compressing nerve roots or other soft tissues. Various intervertebral and interspinous-interlaminar implants have been developed to limit this compression.

Unfortunately, many known treatments are invasive and/or do not provide an adjustable level of distraction between the affected vertebrae. Some implants are difficult or impossible to apply to multiple adjacent vertebral levels without interfering with each other. There is a need for interspinous-interlaminar stabilization systems and methods that overcome these limitations.

SUMMARY

The various systems and methods of the present disclosure have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available interspinous-interlaminar stabilization systems and methods. The systems and methods of the present disclosure may provide interspinous-interlaminar stabilization systems and methods that remedy shortcomings of prior art interspinous-interlaminar stabilization systems and methods.

To achieve the foregoing, and in accordance with the disclosure as embodied and broadly described herein, a system may be provided. The system may be configured to maintain spacing between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina, of adjacent vertebrae of a spine, and may include an implant having an implanted position in which the implant resides in a space between the superior spinous process and superior lamina and the inferior spinous process and inferior lamina. The implant may have a proximal superior surface with a superior concavity shaped to receive

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the superior spinous process in the implanted position, a proximal inferior surface with an inferior concavity shaped to receive the inferior spinous process in the implanted position, a distal superior surface, distal to the proximal superior surface, that faces the superior lamina in the implanted position, a distal inferior surface, distal to the proximal inferior surface, that faces the inferior lamina in the implanted position, and a threaded member extending along a proximal-distal direction. The threaded member may rotatably engage the distal superior surface and the distal inferior surface such that rotation of the threaded member urges the implant to move from a retracted configuration to a deployed configuration by urging the distal superior surface and the distal inferior surface to move apart.

The implant may further have two superior wings extending superiorly from the proximal superior surface such that, in the implanted position, the superior spinous process is received between the superior wings, and two inferior wings extending inferiorly from the proximal superior surface such that, in the implanted position, the inferior spinous process is received between the inferior wings.

The superior wings may have superior tips, and the inferior wings may have inferior tips. The superior tips may be displaced, along the proximal-distal direction, from the inferior tips.

The implant may have a wingless shape. The system may further have a cannula through which the implant is insertable into the space through soft tissues posterior to the space.

The implant may further have a superior member on which the proximal superior surface and the distal superior surface reside, and an inferior member on which the proximal inferior surface and the distal inferior surface reside. The implant may define a cavity, between the superior member and the inferior member, in which at least part of the threaded member resides in the deployed configuration.

The superior member may be shaped to define a superior living hinge that permits the distal superior surface to rotate superiorly, relative to the proximal superior surface, as the implant moves from the retracted configuration to the deployed configuration. The inferior member may be shaped to define an inferior living hinge that permits the distal inferior surface to rotate inferiorly, relative to the proximal inferior surface, as the implant moves from the retracted configuration to the deployed configuration.

The cavity may have a proximal end and a distal end. The implant may further have a threaded block that is operatively connected to the threaded member such that rotation of the threaded member moves the threaded block from the proximal end to the distal end to move the implant from the retracted configuration to the deployed configuration.

The cavity may be shaped such that, in the retracted configuration, the proximal end is wider than the distal end along a superior-inferior direction transverse to the proximal-distal direction. Motion of the threaded block toward the distal end may widen the distal end to urge the distal superior surface and the distal inferior surface to move apart.

The superior member may be formed as a single piece with the inferior member. The implant may further have two laterally-facing surfaces, each of which extends between the superior member and the inferior member and defines an aperture. The apertures may cooperate to define an inserter interface that facilitates coupling of the implant to an inserter.

The distal superior surface and the distal inferior surface may each have a ridge extending along a lateral direction transverse to the proximal-distal direction. The ridges may be positioned to contact the superior spinous process and/or

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the superior lamina and the inferior spinous process and/or the inferior lamina as the implant is moved to the deployed configuration.

The system may further include one or more bone preparation instruments that facilitate preparation of the space for implantation of the implant in the space.

The system may further include an inserter with a proximal end with handle, a distal end configured to grip the implant, and a rotary element configured urge the threaded member to rotate to move the implant from the retracted configuration to the deployed configuration.

A method may be used to implant an implant in a space between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina of adjacent vertebrae of a spine. The method may include, with the implant in a retracted configuration, inserting the implant into the space such that a proximal superior surface of the implant engages the superior spinous process and a proximal inferior surface of the implant engages the inferior spinous process. The method may further include rotating a threaded member extending along a proximal-distal direction to urge the implant to move from the retracted configuration to a deployed configuration by urging a distal superior surface, distal to the proximal superior surface, and a distal inferior surface, distal to the proximal inferior surface, to move apart such that the distal superior surface engages the superior spinous process and/or the superior lamina, and the distal inferior surface engages the inferior spinous process and/or the inferior lamina.

The implant may further include two superior wings extending superiorly from the proximal superior surface, and two inferior wings extending inferiorly from the proximal superior surface. The superior wings may have superior tips. The inferior wings may have inferior tips. The superior tips may be displaced, along the proximal-distal direction, from the inferior tips. Inserting the implant into the space may include causing the superior spinous process to be received between the superior wings, and causing the inferior spinous process to be received between the inferior wings.

The implant may have a wingless shape. Inserting the implant into the space may include inserting the implant through a cannula to pass the implant through soft tissues posterior to the space.

The implant may further have a superior member on which the proximal superior surface and the distal superior surface reside, and an inferior member on which the proximal inferior surface and the distal inferior surface reside. The implant may define a cavity between the superior member and the inferior member. The superior member may be shaped to define a superior living hinge. The inferior member may be shaped to define an inferior living hinge. Urging the implant to move from the retracted configuration to the deployed configuration may include rotating the distal superior surface superiorly, relative to the proximal superior surface via the superior living hinge, and rotating the distal inferior surface inferiorly, relative to the proximal inferior surface, via the inferior living hinge.

The implant may further have a superior member on which the proximal superior surface and the distal superior surface reside, and an inferior member on which the proximal inferior surface and the distal inferior surface reside. The implant may define a cavity between the superior member and the inferior member. The cavity may have a proximal end and a distal end. Urging the implant to move from the retracted configuration to the deployed configuration may include moving a threaded block, in response to

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rotation of the threaded member, from the proximal end to the distal end. The cavity may be shaped such that, in the retracted configuration, the proximal end is wider than the distal end along a superior-inferior direction transverse to the proximal-distal direction. Moving the threaded block toward the distal end may include widening the distal end to urge the distal superior surface and the distal inferior surface to move apart.

The implant may further have a superior member on which the proximal superior surface and the distal superior surface reside, and an inferior member on which the proximal inferior surface and the distal inferior surface reside. The superior member may be formed as a single piece with the inferior member. The implant may further have two laterally-facing surfaces, each of which extends between the superior member and the inferior member and defines an aperture. The apertures may cooperate to define an inserter interface. The method may further include, prior to inserting the implant into the space, coupling the implant to an inserter via the inserter interface, and, after moving the implant from the retracted configuration to a deployed configuration, detaching the implant from the inserter.

The distal superior surface and the distal inferior surface may each have a ridge extending along a lateral direction transverse to the proximal-distal direction. Moving the implant from the retracted configuration to a deployed configuration may include causing the ridges to contact the superior spinous process and/or the superior lamina and the inferior spinous process and/or the inferior lamina.

An implant may be configured, in an implanted position, to reside in a space between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina, of adjacent vertebrae of a spine to maintain spacing between the superior spinous process and the superior lamina, and the inferior spinous process and the inferior lamina. The implant may include a superior member with a proximal superior surface with a superior concavity shaped to receive the superior spinous process in the implanted position, a distal superior surface, distal to the proximal superior surface, that faces the superior lamina in the implanted position. The implant may further include an inferior member with a proximal inferior surface with an inferior concavity shaped to receive the inferior spinous process in the implanted position, a distal inferior surface, distal to the proximal inferior surface, that faces the inferior lamina in the implanted position. The implant may further include a threaded member and a threaded block. The implant may define a cavity between the superior member and the inferior member. The cavity may have a proximal end and a distal end. The threaded member may extend along a proximal-distal direction and rotatably engage the threaded block such that rotation of the threaded member moves the implant from a retracted configuration to a deployed configuration by urging the threaded block to move from the proximal end to the distal end to widen the distal end to urge the distal superior surface and the distal inferior surface to move apart.

These and other features and advantages of the present disclosure will become more fully apparent from the following description and appended claims or may be learned by the practice of the disclosure as set forth hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompa-

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nying drawings. Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the appended claims, the exemplary embodiments of the disclosure will be described with additional specificity and detail through use of the accompanying drawings in which:

FIGS. 1A, 1B, 1C, and 1D are perspective, side elevation, plan, and rear elevation views of an interspinous-interlaminar implant according to one embodiment of the present disclosure, in a retracted configuration.

FIGS. 2A, 2B, 2C, and 2D are perspective, side elevation, plan, and rear elevation views of the implant of FIGS. 1A, 1B, 1C, and 1D, in a deployed configuration.

FIGS. 3A and 3B are perspective views of a probe and a rasp trial suitable for preparing a space between adjacent spinous processes and laminae for implantation of the implant of FIGS. 1A, 1B, 1C, and 1D, according to one embodiment of the present disclosure.

FIG. 4 is a perspective view of an inserter configured to place the implant of FIGS. 1A, 1B, 1C, and 1D, according to one embodiment of the present disclosure.

FIGS. 5A and 5B are perspective views of the implant of FIGS. 1A, 1B, 1C, and 1D implanted in the space between adjacent spinous processes and laminae, in a retracted configuration and a deployed configuration, respectively, according to one embodiment of the present disclosure.

FIG. 6 is a perspective view of a spine with the implant of FIGS. 1A, 1B, 1C, and 1D implanted in four adjacent levels, according to one embodiment of the present disclosure.

FIG. 7 is a perspective view of a spine with the implant of FIGS. 1A, 1B, 1C, and 1D implanted in two adjacent levels in opposite orientations to avoid interference with each other, according to one embodiment of the present disclosure.

FIGS. 8A, 8B, 8C, and 8D are perspective, side elevation, plan, and rear elevation views of an interspinous-interlaminar implant according to another embodiment of the present disclosure, in a retracted configuration.

FIGS. 9A and 9B are perspective views of dilation of soft tissues to access the interspinous process and interlaminar space, and insertion of the implant of FIGS. 8A, 8B, 8C, and 8D in a retracted configuration, respectively, according to one embodiment of the present disclosure.

FIGS. 10A and 10B are perspective views of the implant of FIGS. 8A, 8B, 8C, and 8D in a deployed configuration in the interspinous process and interlaminar space, and a spine with the implant of FIGS. 8A, 8B, 8C, and 8D implanted in four adjacent levels, respectively, according to one embodiment of the present disclosure.

FIGS. 11A and 11B are top plan and side elevation, section views of another interspinous-interlaminar implant according to another embodiment of the present disclosure, in a deployed state.

FIGS. 12A and 12B are top plan and side elevation, section views of yet another interspinous-interlaminar implant according to another embodiment of the present disclosure, in a deployed state.

DETAILED DESCRIPTION

Exemplary embodiments of the disclosure will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the disclosure, as generally described and illustrated in the Figures herein, could be arranged and designed in a wide variety of different

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configurations. Thus, the following more detailed description of the embodiments of the apparatus, system, and method, as represented in FIGS. 1A through 12B, is not intended to limit the scope of the claims, but is merely representative exemplary of exemplary embodiments of the disclosure.

The phrases “connected to,” “coupled to” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be functionally coupled to each other even though they are not in direct contact with each other. The term “abutting” refers to items that are in direct physical contact with each other, although the items may not necessarily be attached together. The phrase “fluid communication” refers to two features that are connected such that a fluid within one feature is able to pass into the other feature.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

The system and method of use in accordance with the present application may overcome one or more of the above-discussed problems commonly associated with conventional interspinous stabilization systems and methods. Specifically, interspinous and interlaminar stabilization systems and methods presented herein may enable interspinous process-interlaminar implants to be reliably placed with smaller incisions, less intrusive implants, and shortened recovery times. These and other unique features of the system and method of use are discussed below and illustrated in the accompanying drawings.

FIGS. 1A, 1B, 1C, and 1D are perspective, side elevation, plan, and rear elevation views of an interspinous-interlaminar implant, or implant 100, according to one embodiment of the present disclosure, in a retracted configuration. The implant 100 may be used to stabilize the range of motion of a superior vertebra 80 relative to an inferior vertebra 82 (shown in FIGS. 5A and 5B). The superior vertebra 80 may have a superior spinous process 90 and a superior lamina 94, and the inferior vertebra 82 may have an inferior spinous process 92 and an inferior lamina 96. The implant 100 may be implanted between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96 to distract the inferior spinous process 92 and the inferior lamina 96 relative to the superior spinous process 90 and the superior lamina 94, thereby limiting posterior compression of nerves and/or other soft tissues between the posterior aspects of the superior vertebra 80 and the inferior vertebra 82. The superior spinous process 90 and the superior lamina 94 may be collectively referred to as the superior spinous process-lamina, and the inferior spinous process 92 and the inferior lamina 96 may be collectively referred to as the inferior spinous process-lamina.

Various directions will be referenced in the present disclosure. These are illustrated in FIG. 1A, and include a superior direction 102, an inferior direction 104, a proximal direction 106, a distal direction 108, and two lateral directions 110 (oriented opposite to each other). The superior direction 102 and the inferior direction 104 may combined, define a superior-inferior direction. The proximal direction 106 and the distal direction 108 may combined, define a

proximal-distal direction. These directions are only shown in FIG. 1A, but apply to all figures and embodiments set forth herein.

As shown, the implant 100 may have a superior member 120, an inferior member 122, and an interconnecting member 124 that couples the superior member 120 to the inferior member 122. The superior member 120 and the inferior member 122 may each be oriented generally along the proximal-distal direction. The superior member 120, the inferior member 122, and the interconnecting member 124 may optionally be formed as a single piece with each other. In alternative embodiments, these members may be formed separately and coupled together through the use of any attachment methods known in the art.

The implant 100 may also have two superior wings 126 extending generally along the superior direction 102 from the superior member 120, and two inferior wings 128 extending generally along the inferior direction 104 from the inferior member 122. When the implant 100 is in position between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96, the superior spinous process 90 may be received between the superior wings 126, and the inferior spinous process 92 may be received between the inferior wings 128. The superior wings 126 and the inferior wings 128 may help keep the implant 100 in place in the space between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96, particularly when the superior vertebra 80 and the inferior vertebra 82 move such that the superior spinous process 90 and the inferior spinous process 92 are drawn apart from each other. The superior wings 126 and the inferior wings 128 may also help further stabilize the superior vertebra 80 relative to the inferior vertebra 82 by lateral motion of the inferior vertebra 82 relative to the superior vertebra 80. However, the superior wings 126 and the inferior wings 128 are optional, and may be omitted in alternative embodiments, as will be discussed subsequently.

The implant 100 may further have a threaded member 130 extending generally along the proximal-distal direction. The threaded member 130 may be operable to move a threaded block 132 distally to move the implant 100 from a retracted configuration to a deployed configuration. In the retracted configuration, shown in FIGS. 1A, 1B, 1C, and 1D, the implant 100 is relatively compact in the superior-inferior direction, and fits relatively easily into the space between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96. Conversely, in the deployed configuration shown in FIGS. 2A, 2B, 2C, and 2D, the distal end of the implant 100 is relatively larger in the superior-inferior direction, causing the superior member 120 and the inferior member 122 to contact and engage the superior spinous process 90 and/or the superior lamina 94 and the inferior spinous process 92 and/or the inferior lamina 96, respectively to restrain motion of the superior spinous process-lamina and the inferior spinous process-lamina toward each other.

The threaded block 132 may move along the proximal-distal direction within a cavity 134 defined between the superior member 120 and the inferior member 122. More specifically, the cavity 134 may have a proximal end 136 and a distal end 137. When the threaded block 132 is in or near the proximal end 136, the implant 100 may be in the retracted configuration. Conversely, when the threaded block 132 is in or near the distal end 137, the implant 100 may be in the deployed configuration. As the threaded block 132 moves towards the distal end, the superior member 120

and inferior member 122 may incrementally deform and deploy, as will be discussed in greater detail subsequently.

The superior member 120 and the inferior member 122 may have windows 138 that provide access to the cavity 134 from superior to the superior member 120, and from inferior to the inferior member 122, respectively. Thus, the windows 138 may receive bony protrusions of the superior spinous process 90, the superior lamina 94, the inferior spinous process 92, and/or the inferior lamina 96 to further secure the implant 100 relative to the superior spinous process-lamina and the inferior spinous process-lamina. Additionally or alternatively, the windows 138 may facilitate osseointegration between the superior spinous process-lamina and the inferior spinous process-lamina. In some embodiments, bone graft or other biologics may be inserted into the cavity 134 before or after implantation of the implant 100 to encourage a column of bone to grow between the superior spinous process-lamina and the inferior spinous process-lamina, through the windows 138 and the cavity 134.

The superior member 120 and the inferior member 122 may each be functionally divided into proximal and distal ends. As embodied in FIGS. 1A, 1B, 1C, and 1D, the proximal and distal ends of the superior member 120 may be formed as a single piece with each other, and the proximal and distal ends of the inferior member 122 may similarly be formed as a single piece with each other. However, in alternative embodiments, a superior member and/or an inferior member may have separately-formed and subsequently coupled proximal and distal portions.

The superior member 120 may have a proximal superior surface 140 and a distal superior surface 142, both of which face toward the superior spinous process 90 and the superior lamina 94. The superior member 120 may also have an interior superior surface 144 that forms the superior boundary of the cavity 134 and faces toward the threaded member 130 and the threaded block 132. The proximal superior surface 140 may have a superior concavity 146 with a concave shape that receives a portion of the superior spinous process 90. The distal superior surface 142 may have superior ridges 148 that extend generally laterally to contact the superior spinous process 90 and/or the superior lamina 94 upon deployment of the implant 100. The superior ridges 148 may have generally sharpened shapes that allow the superior ridges 148 to penetrate the cortical exterior of the superior spinous process 90 and/or the superior lamina 94 to secure the distal superior surface 142 relative to the superior spinous process 90 and the superior lamina 94 upon deployment of the implant 100. In some embodiments, the superior ridges 148 may be positioned to contact only the superior lamina 94.

Similarly, the inferior member 122 may have a proximal inferior surface 150 and a distal inferior surface 152, both of which face toward the inferior spinous process 92 and the inferior lamina 96. The inferior member 122 may also have an interior inferior surface 154 that forms the inferior boundary of the cavity 134 and faces toward the threaded member 130 and the threaded block 132. The proximal inferior surface 150 may have an inferior concavity 156 with a concave shape that receives a portion of the inferior spinous process 92. The distal inferior surface 152 may have inferior ridges 158 that extend generally laterally to contact the inferior spinous process 92 and/or the inferior lamina 96 upon deployment of the implant 100. The inferior ridges 158 may have generally sharpened shapes that allow the inferior ridges 158 to penetrate the cortical exterior of the inferior spinous process 92 and/or the inferior lamina 96 to secure

the distal inferior surface 152 relative to the inferior spinous process 92 and the inferior lamina 96 upon deployment of the implant 100.

The interconnecting member 124 may have a pair of laterally-facing surfaces 160 facing in the lateral directions 110. The laterally-facing surfaces 160 may have a lateral apertures 162 that extend into the member 124 to define an inserter interface that facilitates coupling of the implant 100 to an inserter 400, as will be shown and described in connection with FIG. 4. The interconnecting member 124 may also have a rear aperture 164 that provides access, from a proximal direction, to the threaded member 130 so that the threaded member 130 can be rotated by a user via the inserter 400 to move the implant 100 between the retracted and deployed configurations.

The superior wings 126 may have superior tips 170 that represent the furthest superior extents of the superior wings 126. The superior wings 126 may also have superior teeth 172, which may be positioned proximate the superior tips 170, and protrude inwardly to grip the superior spinous process 90 when the superior spinous process 90 is received between the superior wings 126. The superior teeth 172 may penetrate the cortical exterior of the superior spinous process 90 to secure the superior wings 126 relative to the superior spinous process 90, thereby preventing relative motion between the superior spinous process 90 and the implant 100.

Similarly, the inferior wings 128 may have inferior tips 174 that represent the furthest inferior extents of the inferior wings 128. The inferior wings 128 may also have inferior teeth 176, which may be positioned proximate the inferior tips 174, and protrude inwardly to grip the inferior spinous process 92 when the inferior spinous process 92 is received between the inferior wings 128. The inferior teeth 176 may penetrate the cortical exterior of the inferior spinous process 92 to secure the inferior wings 128 relative to the inferior spinous process 92, thereby preventing relative motion between the inferior spinous process 92 and the implant 100.

Advantageously, the superior tips 170 may be displaced, along the proximal-distal direction, from the inferior tips 174. As shown, the inferior tips 174 may be separated from the superior tips 170 by a superior-inferior displacement 196, and by a proximal-distal displacement 198. The proximal-distal displacement 198 may be sufficient to provide some flexibility whereby the implant 100 can be implanted in the orientation shown in FIG. 1A, or reversed so that the superior wings 126 grip the inferior spinous process 92 and the inferior wings 128 grip the superior spinous process 90. Notably, use of the terms “superior” and “inferior” in relation to the implants described herein does not imply any required orientation upon implantation.

The threaded member 130 may have a head 180 and a shank 182. The head 180 may have an enlarged shape relative to the shank 182. The shank 182 may extend along the proximal-distal direction, as shown in FIGS. 1A, 1B, 1C, and 1D. As further shown, the head 180 may have a socket 184 shaped to receive a corresponding drive feature of the inserter 400. As shown, the socket 184 may have a star shape or the like; this shape may match that of the drive feature so that the drive feature can impart torque to the threaded member 130 to rotate the threaded member 130 relative to the superior member 120 and the inferior member 122. In alternative embodiments, the socket 184 may have a different shape, or may be replaced by one or more positive features such as a boss that cooperate any drive feature shape known in the art.

The shank 182 may have threads 186 with a helical shape that can, upon rotation of the threaded member 130, drive the threaded block 132 to move between the proximal end 136 and the distal end 137 of the cavity 134. A proximal portion of the shank 182 (not shown) may have a smooth portion that passes through the rear aperture 164 of the interconnecting member 124 so that the threaded member 130 is rotatably retained relative to the interconnecting member 124.

The threaded block 132 may have a hole 190, a superior surface 192, and an inferior surface 194. The hole 190 may have threads that receive the threads 186 of the shank 182 of the threaded member 130 such that the threaded member 130 may rotate in place relative to the interconnecting member 124 while driving linear motion of the threaded block 132. The superior surface 192 may engage the interior superior surface 144 of the superior member 120, and the inferior surface 194 may engage the interior inferior surface 154 of the inferior member 122. A snap ring (not visible) or other structure may be used to retain the threaded member 130 relative to the interconnecting member 124.

As shown, the cavity 134 may be shaped such that the proximal end 136 of the cavity 134 is wider, along the superior/inferior direction, than the distal end 137 of the cavity 134. Thus, as the threaded block 132 is driven from the proximal end 136 toward the distal end 137, the superior surface 192 may press against the interior superior surface 144 and the inferior surface 194 may press against the interior inferior surface 154 to widen the distal end 137 of the cavity 134. This may cause the distal portions of the superior member 120 and the inferior member 122 to move apart from each other, thus causing the distal superior surface 142 and the distal inferior surface 152 to move away from each other and toward the superior spinous process-lamina and the inferior spinous process-lamina, respectively. The resulting deployed configuration is shown in FIGS. 2A, 2B, 2C, and 2D.

FIGS. 2A, 2B, 2C, and 2D are perspective, side elevation, plan, and rear elevation views of the implant 100 of FIGS. 1A, 1B, 1C, and 1D, in the deployed configuration. As shown, the threaded block 132 has been driven to reside in the distal end 137 of the cavity 134, and the distal superior surface 142 and the distal inferior surface 152 have been spread apart.

The implant 100 may be formed of any known biocompatible materials, including but not limited to biocompatible metals such as Titanium and Titanium alloys, shape memory alloys such as Nitinol, biocompatible ceramics, and biocompatible polymers such as Polyether ether ketone (PEEK). In some embodiments, the implant 100 may be formed of a less rigid material so that the superior member 120 and the inferior member 122 can bend to spread apart from each other in order to allow the distal end 137 of the cavity 134 to widen as the threaded block 132 moves into the distal end 137.

Further, in some embodiments, the superior member 120 and the inferior member 122 may be made relatively thin so that they can bend between the proximal superior surface 140 and the distal superior surface 142, and between the proximal inferior surface 150 and the distal inferior surface 152. Thus, the superior member 120 may define a superior living hinge 200, and the inferior member 122 may define an inferior living hinge 202. The superior living hinge 200 and the inferior living hinge 202 may enable the distal portions of the superior member 120 and the inferior member 122 to flex outward as the implant 100 moves to the deployed configuration.

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Various instruments may be used to facilitate implantation of the implant 100. Some of these will be shown and described in connection with FIGS. 3A, 3B, and 4. Those of skill in the art will recognize that a wide variety of instruments may be used in connection with the implants disclosed herein, including any instruments known in the art of implanting interspinous process and/or interlaminar devices. Likewise, any surgical methods may be used in connection with the implants disclosed herein, including any surgical methods known in the art of implanting interspinous process and/or interlaminar devices.

FIGS. 3A and 3B are perspective views of a probe 300 and a rasp trial 350 suitable for preparing a space between adjacent spinous processes and laminae for implantation of the implant 100 of FIGS. 1A, 1B, 1C, and 1D, according to one embodiment of the present disclosure. The probe 300 may be used to explore the space between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96, for example, to test for the location of nerves and/or other sensitive tissues. The rasp trial 350 may be used to abrade away excess bone from the adjoining surfaces of the superior spinous process 90, the superior lamina 94, the inferior spinous process 92, and/or the inferior lamina 96 and/or test for proper fit of the implant 100.

More particularly, the probe 300 may have a shaft 310 and a head 320. The shaft 310 may be held by the surgeon, either directly or via another instrument. The head 320 may be inserted into the space between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96. The head 320 may have a blunt tip 330 shaped to facilitate exploration of the space between the superior spinous process-lamina and the inferior spinous process-lamina, while avoiding damage to adjacent soft tissues. The head 320 may further have beveled surfaces 340 that cause the head 320 to have a relatively larger proximal end than its distal end. The beveled surfaces 340 may facilitate insertion of the blunt tip 330 into soft tissues within the space between the superior spinous process-lamina and the inferior spinous process-lamina.

The rasp trial 350 may have a shaft 360 and a head 370. Like the shaft 310 of the probe 300, the shaft 360 may also be held by the surgeon, either directly or with the aid of another instrument. The head 370 may have a blunt tip 380 that also avoids damage to tissues distal to the head 370, which may include nerve tissues such as the spinal column and nerve roots. The head 370 may also have a superior surface 390, an inferior surface 392, and lateral surfaces 394 that are all roughened to permit the head 370 to abrade away surrounding tissues, such as the adjoining surfaces of the superior spinous process-lamina and the inferior spinous process-lamina, in response to reciprocating motion of the head 370 within the space between the superior spinous process-lamina and the inferior spinous process-lamina.

In addition to preparing the space between the superior spinous process 90 and the superior lamina 94, and the inferior spinous process 92 and the inferior lamina 96, the rasp trial 350 may also be used to assess the fit of the implant 100 within the space between the superior spinous process-lamina and the inferior spinous process-lamina. For example, the superior surface 390 may generally mimic the size and shape of the superior aspect of the superior member 120, including the proximal superior surface 140 and the distal superior surface 142. Similarly, the inferior surface 392 may generally mimic the size and shape of the inferior aspect of the inferior member 122, including the proximal inferior surface 150 and the distal inferior surface 152. The

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lateral surfaces 394 may mimic the lateral aspects of the implant 100, including the laterally-facing surfaces 160 of the interconnecting member 124, and the lateral aspects of the superior member 120 and the inferior member 122, combined.

Thus, insertion of the head 370 into the space between the superior spinous process-lamina and the inferior spinous process-lamina may help the surgeon assess whether the implant 100 will fit properly within the space. A kit of implants (not shown) may include implants similar in configuration to the implant 100, but different in size and/or shape, in order to accommodate various bone morphologies. Likewise, the kit may include multiple different rasp trials, each having a size and shape that matches one of the implants of the kit, so that each rasp trial can be used as a trial for one of the implants.

FIG. 4 is a perspective view of an inserter 400 configured to place the implant 100 of FIGS. 1A, 1B, 1C, and 1D, according to one embodiment of the present disclosure. The inserter 400 may also be used to rotate the threaded member 124 to move the implant 100 from the retracted configuration to the deployed configuration.

The inserter 400 may have a body 402, a handle 404, a drive rod 406, and an advancement knob 408. The implant 100 may be coupled to the body 402, and the handle 404 may be gripped by hand to position the implant 100 in the space between the superior spinous process-lamina and the inferior spinous process-lamina. The drive rod 406 may be a rotary element that rotates the threaded member 124, and the advancement knob 408 may selectively advance or retract the drive rod 406.

The body 402 may have a proximal window 410, a shaft 412, and a coupling interface 414. The proximal window 410 may contain the advancement knob 408 such that the advancement knob 408 can be manually rotated by a surgeon. The shaft 412 may extend between the proximal window 410 and the coupling interface 414, which may be positioned at the distal end of the shaft 412.

The coupling interface 414 may have two arms 416 that have bosses 418 that protrude inward. The arms 416 may be spaced apart such that the interconnecting member 124 of the implant 100 can be inserted between the arms 416, with the arms 416 on either lateral side of the interconnecting member 124. The bosses 418 may be sized to be received within the lateral apertures 162 on the laterally-facing surfaces 160 of the interconnecting member 124.

The coupling interface 414 may be configured such that the arms 416 can be flexed apart to permit the bosses 418 to enter and/or exit the lateral apertures 162. Specifically, the shaft 412 may have recesses 420 that permit the arms 416 to flex apart. Thus, the implant 100 may be coupled to the coupling interface 414 by spreading the arms 416 such that the bosses 418 can be inserted into the lateral apertures 162. Likewise, the implant 100 may be decoupled from the coupling interface 414 by again spreading the arms 416 to remove the bosses 418 from the lateral apertures 162.

The drive rod 406 may have a proximal end 430 and a distal end 432. The proximal end 430 may have a proximal flat 434 that can interface with a rotating tool, such as a hand crank or surgical drill. The distal end 432 may have a drive feature 436 that can be coupled to the threaded member 130, for example, by inserting the drive feature 436 into the socket 184 of the head 180 of the threaded member 130. The drive rod 406 may also have a drive knob 438 positioned near the proximal end 430 such that the drive knob 438 can be rotated by hand to rotate the drive rod 406. Thus, the drive rod 406 may be rotated with the proximal flat 434 or the

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drive knob **438**, either with a tool or by hand, to rotate the threaded member **130** to move the implant **100** between the retracted and deployed configurations.

With the implant **100** coupled to the coupling interface **414** of the inserter **400**, the advancement knob **408** may be used to advance the drive feature **436** into the socket **184**, or withdraw the drive feature **436** from the socket **184**. Thus, the drive rod **406** may be disengaged from the implant **100** when the implant **100** has reached the desired level of deployment, without decoupling the inserter **400** from the implant **100**. The advancement knob **408** may optionally have ridges **440**, knurling, or other surface features that facilitate manual gripping and rotation of the advancement knob **408**.

Various surgical methods may be used to place the implant **100**. In some embodiments, the surgical method may include (1) preparation, (2) microsurgical decompression, (3) implant site preparation, (4) implant insertion, and (5) wound closure.

Preparation may include placing the patient in the prone position on a surgical frame, avoiding hyperlordosis of the spinal segment(s) to be operated upon. A neutral position or a slight kyphosis may be advantageous for surgical decompression and/or appropriate interspinous distraction. A mid-line incision may be performed. The muscle may be sharply dissected lateral to the supraspinous ligament, preserving the entire thickness of the supraspinous ligament. Alternatively, the supraspinous ligament may be resected, depending on surgeon's preference. In such a case, the interspinous ligament may be sacrificed and any bony overgrowth of the spinous process that may interfere with insertion may also be resected.

Paraspinal muscles may be stripped off the laminae while preserving the facet capsules. Dependent on the pathology, a microsurgical unilateral decompression may be performed and then the supraspinous ligament together with the fascia and muscle from the opposite side may be mobilized together. The supraspinous ligament may be dissected subperiostally and preserved as a thick cuff and retracted laterally. If possible, a small portion of the bony tip may be resected together with the supraspinous ligament. This may provide for faster healing after reconstruction of the ligament.

Microsurgical decompression may be commenced by resecting the ligamentum flavum. The decompression may then be performed, relieving all points of neural compression. The implant site may be prepared by utilizing trials to define appropriate implant size.

Implant site preparation may be commenced by placing the trial instrument (for example, the rasp trial **350**) to evaluate proper contact with the spinous process and the desired amount of interspinous distraction. The rasp trial **350** may then be used to remove bone to prepare the area for the implant **100**. Some bony resection of the spinous process may be needed to ensure proper contact of the implant **100** with the superior spinous process-lamina and the inferior spinous process-lamina. To ensure the implant **100** is inserted to the proper depth, a small portion of the laminar surface may also be partially resurfaced. Distraction may be appropriate to prevent any settling of the interspinous distance after successful decompression of the spinal stenosis.

Insertion of the implant **100** may be commenced by slightly spreading the superior wings **126** and/or the inferior wings **128** with pliers or another tool, at the mid-portion of each of the superior wings **126** and/or the inferior wings **128**, to ensure the appropriate depth of insertion. The implant **100** may be introduced via impaction utilizing a mallet, for

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example, by striking the proximal end of the inserter **400** with the implant **100** coupled to the inserter **400**. In some embodiments, a handle (not shown) may be secured to the proximal end **430** to facilitate impaction.

The implant **100** may then be expanded by rotating the drive rod **406** clockwise until a tactile click are felt. Proper depth may be determined if a beaded tip probe can be passed freely, leaving a 3-4 mm separation from the dura. If the implant **100** is not seated appropriately, further resurfacing or slightly more impaction force may be utilized. If the superior wings **126** and/or the inferior wings **128** do not have sufficient bony contact, additional stability may be achieved by slightly crimping the superior wings **126** and/or the inferior wings **128**, as appropriate.

Wound closure may be commenced by re-suturing the supraspinous ligament. A "figure 8" suture or the like may be placed through two bone holes in the spinous process and through the supraspinous ligament. Alternatively, the fascia and the supraspinous ligament may be closed in one layer over the spinous processes. A surgical drain may be placed per surgeon preference. Paraspinal muscles may be reattached to the supraspinous ligament. Skin may be closed in the usual manner.

FIGS. **5A** and **5B** are perspective views of the implant **100** of FIGS. **1A**, **1B**, **1C**, and **1D** implanted in the space between adjacent spinous process-laminae, in a retracted configuration and a deployed configuration, respectively, according to one embodiment of the present disclosure. In FIG. **5A**, the implant **100** is retracted to facilitate insertion. The distal superior surface **142** and the distal inferior surface **152** have not yet been spread apart. In FIG. **5B**, the implant **100** has been moved to the deployed configuration to move the distal superior surface **142** and the distal inferior surface **152** into contact with the superior spinous process-lamina and the inferior spinous process-lamina.

Notably, deployment of the implant **100** is optional. In some procedures, the distal superior surface **142** and the distal inferior surface **152** may already be in sufficient contact with the superior spinous process-lamina and the inferior spinous process-lamina after insertion of the implant **100** into the interspinous process-interlaminar space. Thus, deployment of the implant **100** may not be needed.

Further, in some procedures, the implant **100** may be only partially deployed. For example, the distal superior surface **142** and the distal inferior surface **152** may not be in sufficient contact with the superior spinous process-lamina and/or the inferior spinous process-lamina upon insertion of the implant **100** into the interspinous-intralaminar space, but may come into sufficient contact after only partial deployment of the implant **100**. The threaded member **130** may only be rotated sufficiently to move the threaded block **132** to a position between the proximal end **136** and the distal end **137** of the cavity **134**. The distal superior surface **142** and the distal inferior surface **152** may move apart, but not to the extent of their full range of motion.

If desired, the interior superior surface **144** and the interior inferior surface **154** may be shaped to provide generally continuous spreading of the distal superior surface **142** and the distal inferior surface **152** as the threaded member **130** is rotated. The shapes of the interior superior surface **144** and the interior inferior surface **154** may be selected such that the displacement between the distal superior surface **142** and the distal inferior surface **152** varies in linear relationship to rotation of the threaded member **130**. Thus, the motion of the distal superior surface **142** and the distal inferior surface **152** may be relatively gradual and

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predictable, allowing the surgeon to determine the appropriate degree of deployment necessary to obtain the desired level of distraction.

Notably, if the implant **100** and/or corresponding differently-sized implants are to be implanted at multiple levels of a spine, they need not necessarily be deployed to the same degree. The implant size and the appropriate degree of deployment may be made selected for each vertebral level that is to be stabilized.

FIG. **6** is a perspective view of a spine with the implant **100** of FIGS. **1A**, **1B**, **1C**, and **1D** implanted in four adjacent levels of a spine **600**, according to one embodiment of the present disclosure. The implant **100** may be embodied in differing sizes, providing an implant **610**, an implant **620**, an implant **630**, and an implant **640**.

The implant **610**, the implant **620**, and the implant **630** may all closely resemble the implant **100**, with a threaded block **132** that has an elongated cross-sectional shape along the superior-inferior direction. As shown, the implant **610** and the implant **630** are deployed, while the **620** remains in the retracted configuration.

The implant **640** may be smaller than the implant **610**, the implant **620**, and the implant **630**, at least in the superior-inferior direction. Thus, the implant **640** may have a threaded block **632** with a cross-sectional shape that is generally circular, and is thus more compact in the superior-inferior direction than the threaded block **132**. The implant **640** may thus have a lesser degree of expansion upon deployment than the implant **610**, the implant **620**, and the implant **630**. The size and degree of deployment of each implant **610**, **620**, **630**, **640** may be selected to obtain the desired level of distraction, preserving the desired level of lordosis or kyphosis along the length of the spine **600**.

In some instances, where adjacent spinal levels are to be decompressed and stabilized, the implants may be sequentially placed to the appropriate depth that avoids interference of any of the superior wings **126** with any of the inferior wings **128** of adjacent implants. In some embodiments, one or more implants may be flipped vertically to avoid such interference.

FIG. **7** is a perspective view of a spine **700** with the implant **100** of FIGS. **1A**, **1B**, **1C**, and **1D** implanted in two adjacent levels in opposite orientations to avoid interference with each other, according to one embodiment of the present disclosure. The implant **100** may be embodied as an implant **710** and an implant **720** placed at the level immediately below the implant **710**. As shown, the implant **710** may be oriented as shown in FIG. **1A**, while the implant **720** may be turned upside-down so that the inferior wings **128** of the implant **720** are on top, and grip the same spinous process as the inferior wings **128** of the implant **710** above. In this manner, the implant **720** may sufficiently grip the spinous process superior to it without interfering with the inferior wings **128** of the implant **710**.

Notably, presence of the superior wings **126** and the inferior wings **128** is optional. In some embodiments, the superior wings **126** and the inferior wings **128** may be omitted in order to facilitate a less invasive, or even minimally invasive, surgical approach.

FIGS. **8A**, **8B**, **8C**, and **8D** are perspective, side elevation, plan, and rear elevation views of an interspinous-interlaminar implant, or implant **800**, according to another embodiment of the present disclosure, in a retracted configuration. The implant **800** is configured similarly to the implant **100** of FIGS. **1A**, **1B**, **1C**, and **1D**, except that the implant **800** lacks the superior wings **126** and the inferior wings **128**. In place of the superior wings **126** and the inferior wings **128**,

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the implant **800** has superior ridges **826** and inferior ridges **828** that extend generally in the proximal-distal direction.

Like the superior wings **126** and the inferior wings **128**, the superior ridges **826** and the inferior ridges **828** may help to retain the superior spinous process **90** and the inferior spinous process **92** in the superior concavity **146** and the inferior concavity **156**, respectively. If desired, sutures cerclage cables, mechanical fasteners, or other attachment devices may be used to further secure the implant in place between the superior spinous process-lamina and the inferior spinous process-lamina. In some examples, such attachment devices may be passed around or through the superior spinous process **90**, the superior lamina **94**, the inferior spinous process **92** and/or the inferior lamina **96**.

Absence of the superior wings **126** and the inferior wings **128** may make the implant significantly smaller than the implant **100**, so that the implant **800** can be percutaneously delivered to the space between the superior spinous process-lamina and the inferior spinous process-lamina. Thus, some of the exposure and resection steps set forth above for implantation of the implant **100** may not be needed for the implant **800**. Rather, the implant **800** may be delivered through a cannula. Such a cannula may be incorporated into a dilator, such as a sequential dilation system with successively larger cannulas that can be placed in enlarging succession to retract the tissues posterior to the implantation site.

FIGS. **9A** and **9B** are perspective views of dilation of soft tissues to access the interspinous process-interlaminar space of a spine **900**, and insertion of the implant **800** of FIGS. **8A**, **8B**, **8C**, and **8D** in a retracted configuration, respectively, according to one embodiment of the present disclosure. As shown, a dilator **902** may be used to access the surgical site. The dilator **902** may be placed over a K-wire **910**.

As shown in FIG. **9A**, the dilator **902** may have a first cannula **920**, a second cannula **930**, and a third cannula **940**. The first cannula **920** may have a bore sized to receive the K-wire **910**. The second cannula **930** may have a bore sized to receive the first cannula **920**. The third cannula **940** may have a bore sized to receive the second cannula **930**.

Thus, the K-wire **910** may first be anchored in the bone or soft tissue proximate the implantation site. Then, the first cannula **920** may be inserted over the K-wire **910** so that the K-wire **910** is received in the first cannula **920**. The second cannula **930** may be inserted over the first cannula **920** so that the first cannula **920** and the K-wire **910** are received in the second cannula **930**. The third cannula **940** may be inserted over the second cannula **930** so that the second cannula **930**, the first cannula **920**, and the K-wire **910** are received in the third cannula **940**. The first cannula **920**, the second cannula **930**, and the third cannula **940** may each have a tapered tip **950** designed to gently penetrate soft tissues as the cannulas are pressed toward the implantation site.

Once the third cannula **940** has been placed, the second cannula **930**, the first cannula **920**, and the K-wire **910** may be withdrawn, and the third cannula **940** may be used to prepare the implantation site and insert the implant **800**. If desired, any of the steps set forth above in the discussion of surgical methods for the implant **100** may be used. The probe **300** and the rasp **350** may also be used. The inserter **400** used to insert the implant **100** may also be used to install the implant **800** through the third cannula **940** in the retracted configuration. This is shown in FIG. **9B**.

As shown in FIG. **9B**, the implant **800** may be coupled to the inserter **400** as described in connection with the implant **100**. Then, the inserter **400** may be gripped (for example, at

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the handle 404), and maneuvered to insert the implant 800 through the bore of the third cannula 940 and to the implantation site. The implant 800 may be positioned as desired, and optionally, partially, or fully deployed in a manner similar to that of the implant 100.

FIGS. 10A and 10B are perspective views of the implant 800 of FIGS. 8A, 8B, 8C, and 8D in a deployed configuration in the interspinous process-interlaminar space, and a spine 900 with the implant 800 of FIGS. 8A, 8B, 8C, and 8D implanted in four adjacent levels, respectively, according to one embodiment of the present disclosure. Like the implant 100, the implant 800 may be implanted in various sizes and degrees of deployment. FIG. 10A illustrates deployment of the implant 800 inserted through the third cannula 940.

FIG. 10B illustrates the implant 800, embodied as an implant 1010, an implant 1020, an implant 1030, and an implant 1040, used to treat four adjacent levels of the spine 900. For example, the implant 1010, the implant 1020, and the implant 1040 have all been left in the retracted configuration, while the implant 1030 has been deployed. The implant 1040 is of a more compact size, at least in the superior-inferior direction, than the implant 1010, the implant 1020, and the implant 1030.

FIGS. 11A and 11B are top plan and side elevation, section views of another interspinous-interlaminar implant, or implant 1100, according to another embodiment of the present disclosure, in a deployed state. The implant 1100 may be configured substantially the same as the implant 800 of FIGS. 8A, 8B, 8C, and 8D, except that the implant 1100 may have a threaded block 1132 and a cavity 1134 that have been modified to provide tactile and/or auditory confirmation of full deployment, and retain the implant 1100 in the fully deployed configuration.

More specifically, the cavity 1134 may have a proximal end 1136 and a distal end 1137, with an interior superior surface 1144 and an interior inferior surface 1154 that define the superior and inferior boundaries, respectively, of the cavity 1134. The interior superior surface 1144 and the interior inferior surface 1154 may each have a detent feature, such as a bump 1150, that protrudes interiorly.

The threaded block 1132 may have a superior surface 1192 and an inferior surface 1194 that engage the interior superior surface 1144 and the interior inferior surface 1154, respectively, as the threaded block 1132 moves from the proximal end 1136 toward the distal end 1137. The superior surface 1192 and the inferior surface 1194 may each have a detent feature that cooperates with the detent features of the cavity 1134. These detent features may be notches 1140 that receive the bumps 1150 of the superior surface 1192 and the inferior surface 1194 when the threaded block 1132 has moved fully to the distal end 1137, causing full deployment of the implant 1100.

Entry of bumps 1150 into the notches 1140 may be heard and/or felt by the surgeon as an audible and/or tactile “click.” Further, entry of the bumps 1150 into the notches 1140 may provide a dwell point that restricts further motion of the threaded block 1132 along the length of the cavity 1134, thus avoiding over-deployment of the implant 1100.

In alternative embodiments, a wide variety of alternative detent features may be used to provide such auditory and/or tactile feedback and/or such a dwell point. For example, the bumps 1150 and notches 1140 may be reversed so that bumps 1150 are on the threaded block 1132 and notches 1140 are formed in the interior superior surface 1144 and the interior inferior surface 1154. In other embodiments, multiple detent points may be used. For example, with the notches 1140 on the threaded block 1132, multiple sets of

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bumps 1150 may be formed on the interior superior surface 1144 and the interior inferior surface 1154 such that there are multiple “clicks” that occur as the implant 1100 is deployed. If desired, smaller bumps 1150 may be provided proximal to the distal end 1137 of the cavity 1134, such that the corresponding dwell points can be overcome by further actuation of the threaded block 1132, until the threaded block 1132 reaches the distal end 1137.

Further, a snap ring 1195 or other structure may be used to retain the threaded member 1130 relative to the interconnecting member 1124. During assembly, the snap ring 1195 may optionally be inserted into the rear aperture 1164 after insertion of the threaded member 1130 into the rear aperture 1164. The snap ring 1195 may be a split ring, helical ring, or other expandable structure that can be compressed to fit beyond an undercut defined in the rear aperture 1164, and then allowed to expand such that it cannot be withdrawn in the absence of deliberate re-compression, thus preventing inadvertent disassembly of the implant 1100.

FIGS. 12A and 12B are top plan and side elevation, section views of yet another interspinous-interlaminar implant, or implant 1200, according to another embodiment of the present disclosure, in a deployed state. The implant 1200 may be configured substantially the same as the implant 100 of FIGS. 1A, 1B, 1C, and 1D, except that the implant 1200 may have a threaded block 1132 and a cavity 1134 that have been modified to provide tactile and/or auditory confirmation of full deployment, and retain the implant 1200 in the fully deployed configuration. These features may function substantially as described in connection with the implant 1100. Unlike the implant 1100, the implant 1200 may have superior wings 1126 and inferior wings 1128 like those of the implant 100.

Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, Figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.

The phrases “generally parallel” and “generally perpendicular” refer to structures that are within 30° parallelism or perpendicularity relative to each other, respectively. Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the disclosure.

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While specific embodiments and applications of the present disclosure have been illustrated and described, it is to be understood that the disclosure is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present disclosure without departing from its spirit and scope.

What is claimed is:

1. A method for implanting an implant in a space between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina of adjacent vertebrae of a spine, the method comprising:

with the implant in a retracted configuration, inserting the implant along a proximal-to-distal direction into the space such that a proximal superior surface of the implant engages the superior spinous process and a proximal inferior surface of the implant engages the inferior spinous process; and

urging the implant to move from the retracted configuration to a deployed configuration by urging a distal superior surface, distal to the proximal superior surface, and a distal inferior surface, distal to the proximal inferior surface, to move apart such that the distal superior surface engages the superior spinous process and/or the superior lamina and the distal inferior surface engages the inferior spinous process and/or the inferior lamina;

wherein:

the implant comprises a body that is formed as a single piece, the body comprising a superior member and an inferior member that are joined at a proximal end of the body;

the proximal superior surface and the distal superior surface are on the superior member; and

the proximal inferior surface and the distal inferior surface are on the inferior member;

the implant defines a cavity between the superior member and the inferior member;

the superior member is shaped to define a superior living hinge;

the inferior member is shaped to define an inferior living hinge; and

urging the implant to move from the retracted configuration to the deployed configuration comprises:

rotating the distal superior surface superiorly, relative to the proximal superior surface via the superior living hinge; and

rotating the distal inferior surface inferiorly, relative to the proximal inferior surface, via the inferior living hinge.

2. The method of claim 1, wherein:

the implant further comprises:

two superior wings extending superiorly from the proximal superior surface; and

two inferior wings extending inferiorly from the proximal superior surface;

the superior wings have superior tips;

the inferior wings have inferior tips;

the superior tips are displaced, along the proximal-to-distal direction, from the inferior tips; and

inserting the implant into the space comprises:

causing the superior spinous process to be received between the superior wings; and

causing the inferior spinous process to be received between the inferior wings.

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3. The method of claim 1, wherein:

the implant comprises a wingless shape; and

inserting the implant into the space comprises inserting the implant through a cannula to pass the implant through soft tissues posterior to the space.

4. The method of claim 1, wherein:

the implant further comprises a threaded member;

the implant defines a cavity between the superior member and the inferior member;

the cavity comprises a proximal end and a distal end;

urging the implant to move from the retracted configuration to the deployed configuration comprises moving a threaded block, in response to rotation of the threaded member, from the proximal end to the distal end;

the cavity is shaped such that, in the retracted configuration, the proximal end is wider than the distal end along a superior-inferior direction transverse to the proximal-to-distal direction; and

moving the threaded block toward the distal end comprises widening the distal end to urge the distal superior surface and the distal inferior surface to move apart.

5. The method of claim 1, wherein:

the implant further comprises two laterally-facing surfaces, each of which extends between the superior member and the inferior member and defines an aperture;

the apertures cooperate to define an inserter interface; and the method further comprises:

prior to inserting the implant into the space, coupling the implant to an inserter via the inserter interface; and

after moving the implant from the retracted configuration to the deployed configuration, detaching the implant from the inserter.

6. The method of claim 1, wherein:

the distal superior surface and the distal inferior surface each comprises a ridge extending along a lateral direction transverse to the proximal-to-distal direction; and

moving the implant from the retracted configuration to the deployed configuration comprises causing the ridges to contact the superior spinous process and/or the superior lamina and the inferior spinous process and/or the inferior lamina.

7. The method of claim 1, wherein urging the distal superior surface and the distal inferior surface to move apart comprises causing the distal superior surface to engage the superior lamina and causing the distal inferior surface to engage the inferior lamina.

8. A method for implanting an implant in a space between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina of adjacent vertebrae of a spine, the method comprising:

with the implant in a retracted configuration, inserting the implant along a proximal-to-distal direction into the space such that a proximal superior surface of the implant engages the superior spinous process and a proximal inferior surface of the implant engages the inferior spinous process; and

urging the implant to move from the retracted configuration to a deployed configuration by urging a distal superior surface, distal to the proximal superior surface, and a distal inferior surface, distal to the proximal inferior surface, to move apart such that the distal superior surface engages the superior spinous process and/or the superior lamina and the distal inferior surface engages the inferior spinous process and/or the inferior lamina;

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wherein:

the implant comprises a body that is formed as a single piece, the body comprising a superior member and an inferior member;
 the proximal superior surface and the distal superior surface are on the superior member;
 the proximal inferior surface and the distal inferior surface are on the inferior member;
 the superior member and the inferior member each comprise a distal free end; and
 urging the implant to move from the retracted configuration to the deployed configuration comprises causing distal free ends of the superior member and the inferior member to move apart.

9. The method of claim 8, wherein:

the implant further comprises:

two superior wings extending superiorly from the proximal superior surface; and
 two inferior wings extending inferiorly from the proximal superior surface;

the superior wings have superior tips;

the inferior wings have inferior tips;

the superior tips are displaced, along the proximal-to-distal direction, from the inferior tips; and

inserting the implant into the space comprises:

causing the superior spinous process to be received between the superior wings; and

causing the inferior spinous process to be received between the inferior wings.

10. The method of claim 8, wherein:

the implant defines a cavity between the superior member and the inferior member;

the superior member is shaped to define a superior living hinge;

the inferior member is shaped to define an inferior living hinge; and

urging the implant to move from the retracted configuration to the deployed configuration comprises:

rotating the distal superior surface superiorly, relative to the proximal superior surface via the superior living hinge; and

rotating the distal inferior surface inferiorly, relative to the proximal inferior surface, via the inferior living hinge.

11. The method of claim 8, wherein:

the implant further comprises a threaded member;

the implant defines a cavity between the superior member and the inferior member;

the cavity comprises a proximal end and a distal end;

urging the implant to move from the retracted configuration to the deployed configuration comprises moving a threaded block, in response to rotation of the threaded member, from the proximal end to the distal end;

the cavity is shaped such that, in the retracted configuration, the proximal end is wider than the distal end along a superior-inferior direction transverse to the proximal-to-distal direction; and

moving the threaded block toward the distal end comprises widening the distal end to urge the distal superior surface and the distal inferior surface to move apart.

12. The method of claim 8, wherein:

the distal superior surface and the distal inferior surface each comprises a ridge extending along a lateral direction transverse to the proximal-to-distal direction; and
 moving the implant from the retracted configuration to the deployed configuration comprises causing the ridges to

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contact the superior spinous process and/or the superior lamina and the inferior spinous process and/or the inferior lamina.

13. The method of claim 8, wherein urging the distal superior surface and the distal inferior surface to move apart comprises causing the distal superior surface to engage the superior lamina and causing the distal inferior surface to engage the inferior lamina.

14. A method for implanting an implant in a space between a superior spinous process and a superior lamina, and an inferior spinous process and an inferior lamina of adjacent vertebrae of a spine, the method comprising:

with the implant in a retracted configuration, inserting the implant along a proximal-to-distal direction into the space such that a proximal superior surface of the implant engages the superior spinous process and a proximal inferior surface of the implant engages the inferior spinous process; and

urging the implant to move from the retracted configuration to a deployed configuration by urging a distal superior surface, distal to the proximal superior surface, and a distal inferior surface, distal to the proximal inferior surface, to move apart such that the distal superior surface engages the superior lamina and the distal inferior surface engages the inferior lamina;

wherein:

the implant comprises a body that is formed as a single piece, the body comprising a superior member and an inferior member;

the distal superior surface is on the superior member; and

the distal inferior surface is on the inferior member;

the implant defines a cavity between the superior member and the inferior member;

the superior member is shaped to define a superior living hinge;

the inferior member is shaped to define an inferior living hinge; and

urging the implant to move from the retracted configuration to the deployed configuration comprises:

rotating the distal superior surface superiorly, relative to the proximal superior surface via the superior living hinge; and

rotating the distal inferior surface inferiorly, relative to the proximal inferior surface, via the inferior living hinge.

15. The method of claim 14, wherein:

the implant further comprises:

two superior wings extending superiorly from the proximal superior surface; and

two inferior wings extending inferiorly from the proximal superior surface;

the superior wings have superior tips;

the inferior wings have inferior tips;

the superior tips are displaced, along the proximal-to-distal direction, from the inferior tips; and

inserting the implant into the space comprises:

causing the superior spinous process to be received between the superior wings; and

causing the inferior spinous process to be received between the inferior wings.

16. The method of claim 14, wherein:

the implant comprises a wingless shape; and

inserting the implant into the space comprises inserting the implant through a cannula to pass the implant through soft tissues posterior to the space.

17. The method of claim 14, wherein:
the implant further comprises a threaded member;
the implant defines a cavity between the superior member
and the inferior member;
the cavity comprises a proximal end and a distal end; 5
urging the implant to move from the retracted configura-
tion to the deployed configuration comprises moving a
threaded block, in response to rotation of the threaded
member, from the proximal end to the distal end;
the cavity is shaped such that, in the retracted configura- 10
tion, the proximal end is wider than the distal end along
a superior-inferior direction transverse to the proximal-
to-distal direction; and
moving the threaded block toward the distal end com-
prises widening the distal end to urge the distal superior 15
surface and the distal inferior surface to move apart.

18. The method of claim 14, wherein:
the distal superior surface and the distal inferior surface
each comprises a ridge extending along a lateral direc-
tion transverse to the proximal-to-distal direction; and 20
moving the implant from the retracted configuration to the
deployed configuration comprises causing the ridges to
contact the superior spinous process and/or the superior
lamina and the inferior spinous process and/or the
inferior lamina. 25

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